

JPMMethod

Signed Arithmetic

Addition & Subtraction (with Negatives)

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Signed Arithmetic - Addition & Subtraction (with Negatives) Instructions

This workbook helps students become confident in signed number arithmetic, especially when crossing zero. Start simple and work through each section.

- Solve problems slowly and accurately.
- Notice how signs change results.
- Double-check any answers that involve zero.
- Build fluency through consistent repetition.
- Use a number line if needed.
- Try estimating before solving to improve intuition.

Understanding signed arithmetic is essential for progress in math. Keep practicing and enjoy the improvement!

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Section 1. Subtraction: Positive or Slightly Negative Results

Solve each problem and write the answer in the box.

(1) $10 - 9 =$

(7) $9 - 7 =$

(2) $9 - 9 =$

(8) $9 - 9 =$

(3) $7 - 7 =$

(9) $5 - 7 =$

(4) $6 - 6 =$

(10) $1 - 3 =$

(5) $2 - 2 =$

(11) $8 - 6 =$

(6) $5 - 5 =$

(12) $10 - 7 =$

Section 1. Subtraction: Positive or Slightly Negative Results

Solve each problem and write the answer in the box.

(1) $5 - 2 =$

(7) $6 - 5 =$

(2) $8 - 10 =$

(8) $3 - 1 =$

(3) $8 - 3 =$

(9) $1 - 1 =$

(4) $4 - 2 =$

(10) $7 - 7 =$

(5) $7 - 4 =$

(11) $8 - 3 =$

(6) $10 - 8 =$

(12) $7 - 8 =$

Section 1. Subtraction: Positive or Slightly Negative Results

Solve each problem and write the answer in the box.

(1) $5 - 7 =$

(7) $8 - 6 =$

(2) $9 - 6 =$

(8) $10 - 12 =$

(3) $2 - 3 =$

(9) $10 - 4 =$

(4) $7 - 6 =$

(10) $7 - 8 =$

(5) $9 - 4 =$

(11) $3 - 2 =$

(6) $9 - 5 =$

(12) $8 - 4 =$

Section 1. Subtraction: Positive or Slightly Negative Results

Solve each problem and write the answer in the box.

(1) $7 - 7 =$

(7) $3 - 5 =$

(2) $1 - 2 =$

(8) $6 - 7 =$

(3) $5 - 3 =$

(9) $5 - 7 =$

(4) $5 - 6 =$

(10) $10 - 11 =$

(5) $1 - 1 =$

(11) $1 - 3 =$

(6) $10 - 9 =$

(12) $10 - 10 =$

Section 1. Subtraction: Positive or Slightly Negative Results

Solve each problem and write the answer in the box.

(1) $4 - 4 =$

(7) $4 - 3 =$

(2) $6 - 6 =$

(8) $5 - 2 =$

(3) $7 - 8 =$

(9) $9 - 6 =$

(4) $4 - 3 =$

(10) $7 - 9 =$

(5) $9 - 3 =$

(11) $5 - 2 =$

(6) $4 - 2 =$

(12) $10 - 8 =$

Section 1. Subtraction: Positive or Slightly Negative Results

Solve each problem and write the answer in the box.

(1) $1 - 3 =$

(7) $10 - 6 =$

(2) $2 - 2 =$

(8) $8 - 7 =$

(3) $9 - 11 =$

(9) $8 - 1 =$

(4) $9 - 9 =$

(10) $4 - 3 =$

(5) $5 - 2 =$

(11) $8 - 7 =$

(6) $6 - 6 =$

(12) $3 - 1 =$

Section 1. Subtraction: Positive or Slightly Negative Results

Solve each problem and write the answer in the box.

(1) $7 - 7 =$

(7) $7 - 7 =$

(2) $10 - 3 =$

(8) $10 - 2 =$

(3) $2 - 2 =$

(9) $5 - 1 =$

(4) $3 - 3 =$

(10) $10 - 6 =$

(5) $8 - 8 =$

(11) $8 - 2 =$

(6) $9 - 3 =$

(12) $7 - 3 =$

Section 1. Subtraction: Positive or Slightly Negative Results

Solve each problem and write the answer in the box.

(1) $3 - 5 =$

(7) $7 - 1 =$

(2) $2 - 4 =$

(8) $1 - 2 =$

(3) $3 - 5 =$

(9) $8 - 2 =$

(4) $2 - 2 =$

(10) $3 - 5 =$

(5) $9 - 8 =$

(11) $7 - 7 =$

(6) $6 - 6 =$

(12) $5 - 6 =$

Section 1. Subtraction: Positive or Slightly Negative Results

Solve each problem and write the answer in the box.

(1) $5 - 5 =$

(7) $4 - 1 =$

(2) $7 - 7 =$

(8) $2 - 2 =$

(3) $8 - 10 =$

(9) $3 - 3 =$

(4) $10 - 10 =$

(10) $4 - 4 =$

(5) $8 - 10 =$

(11) $6 - 5 =$

(6) $8 - 6 =$

(12) $10 - 2 =$

Section 1. Subtraction: Positive or Slightly Negative Results

Solve each problem and write the answer in the box.

(1) $10 - 11 =$

(7) $4 - 6 =$

(2) $2 - 1 =$

(8) $9 - 3 =$

(3) $5 - 4 =$

(9) $5 - 2 =$

(4) $6 - 7 =$

(10) $10 - 6 =$

(5) $7 - 2 =$

(11) $7 - 1 =$

(6) $7 - 7 =$

(12) $6 - 7 =$

Section 1. Subtraction: Positive or Slightly Negative Results

Solve each problem and write the answer in the box.

(1) $6 - 6 =$

(7) $9 - 3 =$

(2) $1 - 2 =$

(8) $10 - 10 =$

(3) $3 - 4 =$

(9) $5 - 6 =$

(4) $10 - 7 =$

(10) $10 - 9 =$

(5) $5 - 5 =$

(11) $6 - 6 =$

(6) $10 - 6 =$

(12) $8 - 5 =$

Section 1. Subtraction: Positive or Slightly Negative Results

Solve each problem and write the answer in the box.

(1) $6 - 6 =$

(7) $5 - 2 =$

(2) $9 - 1 =$

(8) $4 - 3 =$

(3) $4 - 2 =$

(9) $8 - 9 =$

(4) $7 - 4 =$

(10) $10 - 6 =$

(5) $5 - 1 =$

(11) $9 - 6 =$

(6) $6 - 1 =$

(12) $3 - 4 =$

Section 2. Addition: Negative First Number, Any Result

Solve each problem and write the answer in the box.

(1) $-10 + 4 =$

(7) $-7 + 4 =$

(2) $-15 + 15 =$

(8) $-9 + 10 =$

(3) $-1 + 11 =$

(9) $-3 + 2 =$

(4) $-12 + 7 =$

(10) $-15 + 4 =$

(5) $-13 + 12 =$

(11) $-14 + 10 =$

(6) $-13 + 4 =$

(12) $-13 + 15 =$

Section 2. Addition: Negative First Number, Any Result

Solve each problem and write the answer in the box.

(1) $-6 + 2 =$

(7) $-9 + 1 =$

(2) $-14 + 8 =$

(8) $-10 + 3 =$

(3) $-2 + 13 =$

(9) $-2 + 1 =$

(4) $-4 + 7 =$

(10) $-3 + 2 =$

(5) $-2 + 5 =$

(11) $-6 + 11 =$

(6) $-11 + 10 =$

(12) $-14 + 12 =$

Section 2. Addition: Negative First Number, Any Result

Solve each problem and write the answer in the box.

(1) $-10 + 5 =$

(7) $-4 + 5 =$

(2) $-2 + 15 =$

(8) $-11 + 6 =$

(3) $-4 + 12 =$

(9) $-13 + 3 =$

(4) $-4 + 8 =$

(10) $-12 + 4 =$

(5) $-14 + 15 =$

(11) $-2 + 5 =$

(6) $-3 + 12 =$

(12) $-10 + 10 =$

Section 2. Addition: Negative First Number, Any Result

Solve each problem and write the answer in the box.

(1) $-4 + 5 =$

(7) $-7 + 4 =$

(2) $-15 + 15 =$

(8) $-15 + 9 =$

(3) $-1 + 7 =$

(9) $-5 + 4 =$

(4) $-3 + 5 =$

(10) $-4 + 11 =$

(5) $-10 + 10 =$

(11) $-10 + 5 =$

(6) $-2 + 15 =$

(12) $-3 + 13 =$

Section 2. Addition: Negative First Number, Any Result

Solve each problem and write the answer in the box.

(1) $-9 + 13 =$

(7) $-8 + 15 =$

(2) $-14 + 11 =$

(8) $-5 + 6 =$

(3) $-12 + 15 =$

(9) $-5 + 11 =$

(4) $-9 + 6 =$

(10) $-1 + 8 =$

(5) $-2 + 11 =$

(11) $-10 + 10 =$

(6) $-13 + 8 =$

(12) $-12 + 12 =$

Section 2. Addition: Negative First Number, Any Result

Solve each problem and write the answer in the box.

(1) $-14 + 3 =$

(7) $-3 + 1 =$

(2) $-2 + 2 =$

(8) $-14 + 12 =$

(3) $-2 + 13 =$

(9) $-4 + 2 =$

(4) $-1 + 5 =$

(10) $-1 + 10 =$

(5) $-1 + 2 =$

(11) $-1 + 8 =$

(6) $-6 + 11 =$

(12) $-11 + 6 =$

Section 2. Addition: Negative First Number, Any Result

Solve each problem and write the answer in the box.

(1) $-7 + 10 =$

(7) $-9 + 15 =$

(2) $-5 + 14 =$

(8) $-5 + 2 =$

(3) $-6 + 12 =$

(9) $-9 + 1 =$

(4) $-12 + 7 =$

(10) $-12 + 4 =$

(5) $-3 + 14 =$

(11) $-13 + 1 =$

(6) $-11 + 11 =$

(12) $-2 + 5 =$

Section 2. Addition: Negative First Number, Any Result

Solve each problem and write the answer in the box.

(1) $-1 + 13 =$

(7) $-5 + 5 =$

(2) $-1 + 7 =$

(8) $-14 + 5 =$

(3) $-14 + 6 =$

(9) $-10 + 13 =$

(4) $-6 + 6 =$

(10) $-11 + 8 =$

(5) $-10 + 8 =$

(11) $-15 + 2 =$

(6) $-12 + 15 =$

(12) $-5 + 4 =$

Section 2. Addition: Negative First Number, Any Result

Solve each problem and write the answer in the box.

(1) $-9 + 4 =$

(7) $-1 + 1 =$

(2) $-5 + 10 =$

(8) $-4 + 13 =$

(3) $-14 + 9 =$

(9) $-12 + 12 =$

(4) $-11 + 4 =$

(10) $-3 + 11 =$

(5) $-6 + 9 =$

(11) $-3 + 3 =$

(6) $-1 + 2 =$

(12) $-14 + 1 =$

Section 2. Addition: Negative First Number, Any Result

Solve each problem and write the answer in the box.

(1) $-10 + 8 =$

(7) $-11 + 12 =$

(2) $-3 + 8 =$

(8) $-7 + 2 =$

(3) $-5 + 7 =$

(9) $-11 + 2 =$

(4) $-7 + 4 =$

(10) $-11 + 15 =$

(5) $-12 + 12 =$

(11) $-5 + 12 =$

(6) $-1 + 10 =$

(12) $-10 + 1 =$

Section 2. Addition: Negative First Number, Any Result

Solve each problem and write the answer in the box.

(1) $-3 + 13 =$

(7) $-2 + 13 =$

(2) $-7 + 11 =$

(8) $-3 + 9 =$

(3) $-7 + 7 =$

(9) $-9 + 6 =$

(4) $-14 + 15 =$

(10) $-7 + 14 =$

(5) $-4 + 11 =$

(11) $-12 + 4 =$

(6) $-13 + 8 =$

(12) $-6 + 5 =$

Section 2. Addition: Negative First Number, Any Result

Solve each problem and write the answer in the box.

(1) $-3 + 15 =$

(7) $-4 + 10 =$

(2) $-14 + 9 =$

(8) $-10 + 6 =$

(3) $-2 + 12 =$

(9) $-4 + 15 =$

(4) $-11 + 7 =$

(10) $-5 + 14 =$

(5) $-14 + 2 =$

(11) $-10 + 3 =$

(6) $-11 + 4 =$

(12) $-6 + 8 =$

Section 3. Simple Subtraction (Mixed Results)

Solve each problem and write the answer in the box.

(1) $2 - 2 =$

(7) $3 - 5 =$

(2) $4 - 6 =$

(8) $1 - 2 =$

(3) $2 - 3 =$

(9) $2 - 6 =$

(4) $4 - 2 =$

(10) $5 - 5 =$

(5) $7 - 5 =$

(11) $3 - 8 =$

(6) $3 - 3 =$

(12) $7 - 6 =$

Section 3. Simple Subtraction (Mixed Results)

Solve each problem and write the answer in the box.

(1) $8 - 2 =$

(7) $3 - 2 =$

(2) $4 - 1 =$

(8) $6 - 3 =$

(3) $7 - 3 =$

(9) $4 - 7 =$

(4) $1 - 5 =$

(10) $8 - 6 =$

(5) $5 - 5 =$

(11) $4 - 5 =$

(6) $5 - 4 =$

(12) $2 - 2 =$

Section 3. Simple Subtraction (Mixed Results)

Solve each problem and write the answer in the box.

(1) $3 - 4 =$

(7) $3 - 1 =$

(2) $7 - 3 =$

(8) $7 - 8 =$

(3) $5 - 3 =$

(9) $4 - 1 =$

(4) $7 - 6 =$

(10) $5 - 7 =$

(5) $8 - 7 =$

(11) $7 - 5 =$

(6) $3 - 7 =$

(12) $8 - 1 =$

Section 3. Simple Subtraction (Mixed Results)

Solve each problem and write the answer in the box.

(1) $7 - 2 =$

(7) $5 - 1 =$

(2) $8 - 8 =$

(8) $1 - 7 =$

(3) $3 - 7 =$

(9) $2 - 2 =$

(4) $3 - 5 =$

(10) $6 - 1 =$

(5) $7 - 6 =$

(11) $1 - 6 =$

(6) $6 - 6 =$

(12) $1 - 4 =$

Section 3. Simple Subtraction (Mixed Results)

Solve each problem and write the answer in the box.

(1) $1 - 1 =$

(7) $8 - 2 =$

(2) $1 - 6 =$

(8) $3 - 1 =$

(3) $7 - 1 =$

(9) $6 - 3 =$

(4) $6 - 7 =$

(10) $6 - 5 =$

(5) $2 - 6 =$

(11) $7 - 2 =$

(6) $6 - 2 =$

(12) $2 - 5 =$

Section 3. Simple Subtraction (Mixed Results)

Solve each problem and write the answer in the box.

(1) $6 - 4 =$

(7) $8 - 6 =$

(2) $7 - 1 =$

(8) $1 - 5 =$

(3) $4 - 4 =$

(9) $3 - 5 =$

(4) $1 - 3 =$

(10) $1 - 8 =$

(5) $7 - 5 =$

(11) $2 - 5 =$

(6) $7 - 6 =$

(12) $1 - 2 =$

Section 3. Simple Subtraction (Mixed Results)

Solve each problem and write the answer in the box.

(1) $8 - 1 =$

(7) $7 - 1 =$

(2) $6 - 4 =$

(8) $1 - 6 =$

(3) $3 - 3 =$

(9) $7 - 7 =$

(4) $7 - 4 =$

(10) $1 - 7 =$

(5) $4 - 8 =$

(11) $7 - 6 =$

(6) $8 - 6 =$

(12) $2 - 7 =$

Section 3. Simple Subtraction (Mixed Results)

Solve each problem and write the answer in the box.

(1) $4 - 4 =$

(7) $2 - 5 =$

(2) $6 - 8 =$

(8) $7 - 8 =$

(3) $6 - 3 =$

(9) $1 - 3 =$

(4) $2 - 4 =$

(10) $2 - 1 =$

(5) $1 - 4 =$

(11) $5 - 1 =$

(6) $5 - 4 =$

(12) $5 - 6 =$

Section 3. Simple Subtraction (Mixed Results)

Solve each problem and write the answer in the box.

(1) $5 - 1 =$

(7) $5 - 8 =$

(2) $6 - 3 =$

(8) $3 - 2 =$

(3) $3 - 7 =$

(9) $8 - 1 =$

(4) $2 - 8 =$

(10) $7 - 4 =$

(5) $3 - 5 =$

(11) $8 - 5 =$

(6) $8 - 2 =$

(12) $1 - 3 =$

Section 3. Simple Subtraction (Mixed Results)

Solve each problem and write the answer in the box.

(1) $5 - 1 =$

(7) $2 - 5 =$

(2) $6 - 8 =$

(8) $7 - 6 =$

(3) $6 - 4 =$

(9) $4 - 8 =$

(4) $5 - 7 =$

(10) $3 - 2 =$

(5) $5 - 2 =$

(11) $3 - 3 =$

(6) $2 - 4 =$

(12) $1 - 1 =$

Section 3. Simple Subtraction (Mixed Results)

Solve each problem and write the answer in the box.

(1) $1 - 4 =$

(7) $1 - 7 =$

(2) $4 - 4 =$

(8) $3 - 6 =$

(3) $5 - 5 =$

(9) $4 - 2 =$

(4) $1 - 3 =$

(10) $2 - 3 =$

(5) $6 - 3 =$

(11) $8 - 1 =$

(6) $2 - 4 =$

(12) $2 - 1 =$

Section 3. Simple Subtraction (Mixed Results)

Solve each problem and write the answer in the box.

(1) $3 - 1 =$

(7) $5 - 4 =$

(2) $1 - 5 =$

(8) $4 - 7 =$

(3) $6 - 7 =$

(9) $6 - 2 =$

(4) $6 - 3 =$

(10) $8 - 6 =$

(5) $7 - 8 =$

(11) $4 - 8 =$

(6) $3 - 5 =$

(12) $1 - 1 =$

Section 4. Subtraction up to 20 (Mixed Results)

Solve each problem and write the answer in the box.

(1) $11 - 15 =$

(7) $8 - 9 =$

(2) $11 - 8 =$

(8) $5 - 5 =$

(3) $9 - 7 =$

(9) $12 - 15 =$

(4) $6 - 14 =$

(10) $15 - 12 =$

(5) $11 - 7 =$

(11) $11 - 6 =$

(6) $10 - 5 =$

(12) $7 - 10 =$

Section 4. Subtraction up to 20 (Mixed Results)

Solve each problem and write the answer in the box.

(1) $6 - 11 =$

(7) $14 - 7 =$

(2) $11 - 15 =$

(8) $8 - 8 =$

(3) $5 - 14 =$

(9) $14 - 15 =$

(4) $5 - 5 =$

(10) $11 - 8 =$

(5) $12 - 8 =$

(11) $7 - 9 =$

(6) $12 - 13 =$

(12) $5 - 13 =$

Section 4. Subtraction up to 20 (Mixed Results)

Solve each problem and write the answer in the box.

(1) $8 - 11 =$

(7) $13 - 5 =$

(2) $5 - 7 =$

(8) $6 - 11 =$

(3) $15 - 9 =$

(9) $13 - 10 =$

(4) $7 - 14 =$

(10) $10 - 9 =$

(5) $15 - 7 =$

(11) $6 - 7 =$

(6) $5 - 6 =$

(12) $11 - 6 =$

Section 4. Subtraction up to 20 (Mixed Results)

Solve each problem and write the answer in the box.

(1) $5 - 5 =$

(7) $11 - 13 =$

(2) $7 - 11 =$

(8) $6 - 7 =$

(3) $13 - 8 =$

(9) $15 - 10 =$

(4) $15 - 6 =$

(10) $8 - 8 =$

(5) $13 - 14 =$

(11) $5 - 11 =$

(6) $12 - 11 =$

(12) $10 - 12 =$

Section 4. Subtraction up to 20 (Mixed Results)

Solve each problem and write the answer in the box.

(1) $15 - 15 =$

(7) $11 - 5 =$

(2) $9 - 12 =$

(8) $7 - 13 =$

(3) $12 - 9 =$

(9) $7 - 5 =$

(4) $11 - 8 =$

(10) $14 - 7 =$

(5) $12 - 8 =$

(11) $14 - 13 =$

(6) $13 - 8 =$

(12) $11 - 7 =$

Section 4. Subtraction up to 20 (Mixed Results)

Solve each problem and write the answer in the box.

(1) $10 - 14 =$

(7) $8 - 7 =$

(2) $5 - 10 =$

(8) $12 - 12 =$

(3) $5 - 5 =$

(9) $14 - 11 =$

(4) $10 - 7 =$

(10) $10 - 8 =$

(5) $6 - 8 =$

(11) $8 - 10 =$

(6) $15 - 14 =$

(12) $9 - 11 =$

Section 4. Subtraction up to 20 (Mixed Results)

Solve each problem and write the answer in the box.

(1) $15 - 15 =$

(7) $8 - 11 =$

(2) $10 - 8 =$

(8) $6 - 14 =$

(3) $11 - 14 =$

(9) $7 - 9 =$

(4) $6 - 5 =$

(10) $9 - 14 =$

(5) $15 - 9 =$

(11) $8 - 9 =$

(6) $13 - 5 =$

(12) $11 - 12 =$

Section 4. Subtraction up to 20 (Mixed Results)

Solve each problem and write the answer in the box.

(1) $15 - 8 =$

(7) $10 - 13 =$

(2) $6 - 6 =$

(8) $12 - 13 =$

(3) $13 - 11 =$

(9) $8 - 11 =$

(4) $9 - 12 =$

(10) $7 - 7 =$

(5) $11 - 10 =$

(11) $5 - 8 =$

(6) $7 - 6 =$

(12) $10 - 10 =$

Section 4. Subtraction up to 20 (Mixed Results)

Solve each problem and write the answer in the box.

(1) $7 - 5 =$

(7) $15 - 8 =$

(2) $14 - 8 =$

(8) $8 - 15 =$

(3) $14 - 12 =$

(9) $6 - 6 =$

(4) $9 - 14 =$

(10) $7 - 12 =$

(5) $5 - 12 =$

(11) $11 - 12 =$

(6) $13 - 12 =$

(12) $11 - 5 =$

Section 4. Subtraction up to 20 (Mixed Results)

Solve each problem and write the answer in the box.

(1) $6 - 11 =$

(7) $7 - 5 =$

(2) $5 - 8 =$

(8) $13 - 5 =$

(3) $6 - 15 =$

(9) $5 - 9 =$

(4) $13 - 15 =$

(10) $6 - 7 =$

(5) $8 - 5 =$

(11) $14 - 15 =$

(6) $12 - 14 =$

(12) $10 - 12 =$

Section 4. Subtraction up to 20 (Mixed Results)

Solve each problem and write the answer in the box.

(1) $7 - 6 =$

(7) $9 - 14 =$

(2) $13 - 10 =$

(8) $14 - 13 =$

(3) $9 - 8 =$

(9) $5 - 7 =$

(4) $9 - 12 =$

(10) $11 - 11 =$

(5) $7 - 5 =$

(11) $14 - 5 =$

(6) $12 - 14 =$

(12) $14 - 12 =$

Section 4. Subtraction up to 20 (Mixed Results)

Solve each problem and write the answer in the box.

(1) $10 - 11 =$

(7) $15 - 11 =$

(2) $12 - 10 =$

(8) $7 - 5 =$

(3) $15 - 10 =$

(9) $9 - 15 =$

(4) $11 - 12 =$

(10) $12 - 7 =$

(5) $7 - 6 =$

(11) $9 - 13 =$

(6) $11 - 9 =$

(12) $13 - 9 =$

Section 5. Mixed Positive Addition & Subtraction

Solve each problem and write the answer in the box.

(1) $4 + 5 =$

(7) $2 - 3 =$

(2) $10 - 1 =$

(8) $5 - 7 =$

(3) $8 - 4 =$

(9) $9 - 9 =$

(4) $5 + 5 =$

(10) $6 - 7 =$

(5) $8 - 6 =$

(11) $5 - 9 =$

(6) $1 + 7 =$

(12) $1 - 5 =$

Section 5. Mixed Positive Addition & Subtraction

Solve each problem and write the answer in the box.

(1) $5 - 4 =$

(7) $9 + 2 =$

(2) $2 + 4 =$

(8) $5 - 10 =$

(3) $8 + 1 =$

(9) $2 - 10 =$

(4) $4 - 8 =$

(10) $1 + 4 =$

(5) $10 + 3 =$

(11) $6 + 8 =$

(6) $2 - 2 =$

(12) $10 + 1 =$

Section 5. Mixed Positive Addition & Subtraction

Solve each problem and write the answer in the box.

(1) $6 - 8 =$

(7) $3 + 7 =$

(2) $3 + 3 =$

(8) $4 + 2 =$

(3) $4 + 1 =$

(9) $5 - 4 =$

(4) $10 - 9 =$

(10) $9 + 1 =$

(5) $9 - 8 =$

(11) $8 - 7 =$

(6) $10 + 4 =$

(12) $6 - 1 =$

Section 5. Mixed Positive Addition & Subtraction

Solve each problem and write the answer in the box.

(1) $8 + 7 =$

(7) $3 + 1 =$

(2) $8 + 2 =$

(8) $10 + 8 =$

(3) $3 + 5 =$

(9) $1 + 9 =$

(4) $7 + 10 =$

(10) $5 - 1 =$

(5) $6 + 6 =$

(11) $4 - 5 =$

(6) $5 - 8 =$

(12) $1 + 6 =$

Section 5. Mixed Positive Addition & Subtraction

Solve each problem and write the answer in the box.

(1) $5 - 1 =$

(7) $5 + 8 =$

(2) $9 + 2 =$

(8) $4 + 1 =$

(3) $2 + 8 =$

(9) $8 - 6 =$

(4) $5 + 1 =$

(10) $7 + 2 =$

(5) $1 + 7 =$

(11) $7 - 3 =$

(6) $8 + 4 =$

(12) $5 - 7 =$

Section 5. Mixed Positive Addition & Subtraction

Solve each problem and write the answer in the box.

(1) $5 - 3 =$

(7) $1 + 10 =$

(2) $7 - 10 =$

(8) $7 - 9 =$

(3) $6 + 2 =$

(9) $8 + 9 =$

(4) $8 + 2 =$

(10) $3 - 9 =$

(5) $4 + 9 =$

(11) $2 + 3 =$

(6) $6 + 7 =$

(12) $1 + 3 =$

Section 5. Mixed Positive Addition & Subtraction

Solve each problem and write the answer in the box.

(1) $5 - 8 =$

(7) $6 + 5 =$

(2) $1 + 3 =$

(8) $6 - 3 =$

(3) $8 + 3 =$

(9) $1 - 4 =$

(4) $5 - 6 =$

(10) $3 + 6 =$

(5) $10 - 10 =$

(11) $8 + 2 =$

(6) $9 - 7 =$

(12) $6 + 1 =$

Section 5. Mixed Positive Addition & Subtraction

Solve each problem and write the answer in the box.

(1) $6 - 3 =$

(7) $8 + 10 =$

(2) $4 - 5 =$

(8) $1 + 7 =$

(3) $5 + 3 =$

(9) $1 - 3 =$

(4) $3 + 8 =$

(10) $5 - 5 =$

(5) $6 + 2 =$

(11) $6 + 9 =$

(6) $8 - 3 =$

(12) $8 - 9 =$

Section 5. Mixed Positive Addition & Subtraction

Solve each problem and write the answer in the box.

(1) $6 - 9 =$

(7) $7 + 8 =$

(2) $10 - 1 =$

(8) $2 - 7 =$

(3) $3 + 5 =$

(9) $9 + 2 =$

(4) $2 + 8 =$

(10) $4 + 8 =$

(5) $6 + 8 =$

(11) $10 - 2 =$

(6) $1 + 5 =$

(12) $9 + 5 =$

Section 5. Mixed Positive Addition & Subtraction

Solve each problem and write the answer in the box.

(1) $1 + 9 =$

(7) $4 - 1 =$

(2) $7 + 7 =$

(8) $9 + 2 =$

(3) $2 + 5 =$

(9) $8 + 9 =$

(4) $8 - 3 =$

(10) $3 + 6 =$

(5) $7 + 3 =$

(11) $2 - 8 =$

(6) $10 - 5 =$

(12) $8 + 4 =$

Section 5. Mixed Positive Addition & Subtraction

Solve each problem and write the answer in the box.

(1) $5 + 1 =$

(7) $10 - 10 =$

(2) $3 + 10 =$

(8) $9 + 7 =$

(3) $6 + 1 =$

(9) $7 - 10 =$

(4) $5 - 1 =$

(10) $5 - 3 =$

(5) $3 - 5 =$

(11) $8 - 6 =$

(6) $9 + 6 =$

(12) $1 + 4 =$

Section 5. Mixed Positive Addition & Subtraction

Solve each problem and write the answer in the box.

(1) $9 + 8 =$

(7) $7 - 6 =$

(2) $4 + 9 =$

(8) $8 - 10 =$

(3) $10 + 5 =$

(9) $9 + 3 =$

(4) $4 - 7 =$

(10) $8 + 6 =$

(5) $4 - 2 =$

(11) $9 - 6 =$

(6) $4 + 1 =$

(12) $1 + 1 =$

Section 6. Negative Second Term (Addition)

Solve each problem and write the answer in the box.

(1) $8 - 8 =$

(7) $9 - 10 =$

(2) $4 - 2 =$

(8) $3 - 6 =$

(3) $9 - 2 =$

(9) $6 - 8 =$

(4) $10 - 10 =$

(10) $2 - 7 =$

(5) $4 - 11 =$

(11) $3 - 14 =$

(6) $2 - 10 =$

(12) $1 - 10 =$

Section 6. Negative Second Term (Addition)

Solve each problem and write the answer in the box.

(1) $9 - 15 =$

(7) $2 - 11 =$

(2) $4 - 2 =$

(8) $1 - 11 =$

(3) $3 - 2 =$

(9) $7 - 10 =$

(4) $3 - 4 =$

(10) $2 - 1 =$

(5) $6 - 4 =$

(11) $2 - 14 =$

(6) $4 - 14 =$

(12) $10 - 6 =$

Section 6. Negative Second Term (Addition)

Solve each problem and write the answer in the box.

(1) $8 - 1 =$

(7) $10 - 9 =$

(2) $2 - 3 =$

(8) $2 - 13 =$

(3) $9 - 3 =$

(9) $7 - 13 =$

(4) $7 - 14 =$

(10) $3 - 14 =$

(5) $1 - 5 =$

(11) $8 - 4 =$

(6) $2 - 12 =$

(12) $9 - 8 =$

Section 6. Negative Second Term (Addition)

Solve each problem and write the answer in the box.

(1) $2 - 12 =$

(7) $8 - 14 =$

(2) $5 - 5 =$

(8) $2 - 8 =$

(3) $10 - 9 =$

(9) $4 - 13 =$

(4) $7 - 9 =$

(10) $5 - 12 =$

(5) $1 - 12 =$

(11) $9 - 7 =$

(6) $10 - 2 =$

(12) $3 - 4 =$

Section 6. Negative Second Term (Addition)

Solve each problem and write the answer in the box.

(1) $1 - 4 =$

(7) $3 - 11 =$

(2) $1 - 2 =$

(8) $4 - 9 =$

(3) $3 - 9 =$

(9) $5 - 6 =$

(4) $6 - 15 =$

(10) $3 - 10 =$

(5) $7 - 4 =$

(11) $9 - 10 =$

(6) $7 - 2 =$

(12) $7 - 15 =$

Section 6. Negative Second Term (Addition)

Solve each problem and write the answer in the box.

(1) $2 - 14 =$

(7) $2 - 15 =$

(2) $1 - 9 =$

(8) $8 - 9 =$

(3) $2 - 6 =$

(9) $7 - 1 =$

(4) $6 - 1 =$

(10) $10 - 2 =$

(5) $1 - 10 =$

(11) $7 - 9 =$

(6) $3 - 4 =$

(12) $7 - 11 =$

Section 6. Negative Second Term (Addition)

Solve each problem and write the answer in the box.

(1) $1 - 12 =$

(7) $4 - 11 =$

(2) $1 - 1 =$

(8) $8 - 12 =$

(3) $8 - 9 =$

(9) $6 - 4 =$

(4) $7 - 4 =$

(10) $7 - 1 =$

(5) $3 - 8 =$

(11) $9 - 13 =$

(6) $7 - 2 =$

(12) $6 - 12 =$

Section 6. Negative Second Term (Addition)

Solve each problem and write the answer in the box.

(1) $3 - 10 =$

(7) $10 - 7 =$

(2) $2 - 15 =$

(8) $6 - 10 =$

(3) $10 - 3 =$

(9) $8 - 2 =$

(4) $4 - 2 =$

(10) $10 - 9 =$

(5) $8 - 6 =$

(11) $2 - 10 =$

(6) $7 - 6 =$

(12) $5 - 5 =$

Section 6. Negative Second Term (Addition)

Solve each problem and write the answer in the box.

(1) $8 - 10 =$

(7) $9 - 7 =$

(2) $5 - 13 =$

(8) $10 - 13 =$

(3) $4 - 6 =$

(9) $5 - 11 =$

(4) $1 - 10 =$

(10) $9 - 9 =$

(5) $5 - 15 =$

(11) $8 - 2 =$

(6) $2 - 6 =$

(12) $1 - 3 =$

Section 6. Negative Second Term (Addition)

Solve each problem and write the answer in the box.

(1) $1 - 11 =$

(7) $6 - 5 =$

(2) $7 - 13 =$

(8) $9 - 3 =$

(3) $3 - 10 =$

(9) $6 - 4 =$

(4) $2 - 6 =$

(10) $7 - 5 =$

(5) $2 - 1 =$

(11) $2 - 9 =$

(6) $2 - 8 =$

(12) $7 - 12 =$

Section 6. Negative Second Term (Addition)

Solve each problem and write the answer in the box.

(1) $9 - 11 =$

(7) $8 - 11 =$

(2) $3 - 3 =$

(8) $2 - 8 =$

(3) $10 - 15 =$

(9) $4 - 5 =$

(4) $6 - 4 =$

(10) $3 - 8 =$

(5) $1 - 13 =$

(11) $6 - 6 =$

(6) $5 - 2 =$

(12) $6 - 5 =$

Section 6. Negative Second Term (Addition)

Solve each problem and write the answer in the box.

(1) $10 - 2 =$

(7) $8 - 2 =$

(2) $4 - 14 =$

(8) $7 - 13 =$

(3) $2 - 6 =$

(9) $2 - 11 =$

(4) $9 - 13 =$

(10) $5 - 2 =$

(5) $1 - 2 =$

(11) $6 - 6 =$

(6) $9 - 10 =$

(12) $4 - 12 =$

Section 7. Negative First Term (Addition)

Solve each problem and write the answer in the box.

(1) $-2 + 5 =$

(7) $-14 + 6 =$

(2) $-15 + 1 =$

(8) $-1 + 11 =$

(3) $-11 + 6 =$

(9) $-9 + 1 =$

(4) $-13 + 5 =$

(10) $-9 + 5 =$

(5) $-7 + 9 =$

(11) $-7 + 2 =$

(6) $-2 + 8 =$

(12) $-2 + 11 =$

Section 7. Negative First Term (Addition)

Solve each problem and write the answer in the box.

(1) $-13 + 11 =$

(7) $-1 + 14 =$

(2) $-8 + 5 =$

(8) $-8 + 14 =$

(3) $-14 + 6 =$

(9) $-9 + 2 =$

(4) $-10 + 6 =$

(10) $-15 + 12 =$

(5) $-7 + 9 =$

(11) $-9 + 4 =$

(6) $-1 + 10 =$

(12) $-14 + 5 =$

Section 7. Negative First Term (Addition)

Solve each problem and write the answer in the box.

(1) $-3 + 2 =$

(7) $-11 + 4 =$

(2) $-1 + 5 =$

(8) $-15 + 2 =$

(3) $-2 + 9 =$

(9) $-2 + 4 =$

(4) $-11 + 6 =$

(10) $-13 + 5 =$

(5) $-12 + 11 =$

(11) $-11 + 9 =$

(6) $-8 + 9 =$

(12) $-7 + 7 =$

Section 7. Negative First Term (Addition)

Solve each problem and write the answer in the box.

(1) $-8 + 5 =$

(7) $-1 + 3 =$

(2) $-9 + 13 =$

(8) $-1 + 12 =$

(3) $-5 + 1 =$

(9) $-1 + 4 =$

(4) $-2 + 13 =$

(10) $-1 + 2 =$

(5) $-12 + 1 =$

(11) $-6 + 1 =$

(6) $-6 + 4 =$

(12) $-15 + 12 =$

Section 7. Negative First Term (Addition)

Solve each problem and write the answer in the box.

(1) $-14 + 5 =$

(7) $-15 + 9 =$

(2) $-8 + 4 =$

(8) $-7 + 14 =$

(3) $-1 + 1 =$

(9) $-4 + 14 =$

(4) $-10 + 3 =$

(10) $-6 + 9 =$

(5) $-3 + 5 =$

(11) $-12 + 5 =$

(6) $-7 + 8 =$

(12) $-8 + 8 =$

Section 7. Negative First Term (Addition)

Solve each problem and write the answer in the box.

(1) $-6 + 11 =$

(7) $-4 + 6 =$

(2) $-5 + 8 =$

(8) $-10 + 15 =$

(3) $-1 + 4 =$

(9) $-7 + 8 =$

(4) $-12 + 11 =$

(10) $-14 + 9 =$

(5) $-9 + 9 =$

(11) $-13 + 12 =$

(6) $-12 + 1 =$

(12) $-9 + 6 =$

Section 7. Negative First Term (Addition)

Solve each problem and write the answer in the box.

(1) $-4 + 11 =$

(7) $-7 + 14 =$

(2) $-9 + 8 =$

(8) $-15 + 8 =$

(3) $-14 + 13 =$

(9) $-14 + 12 =$

(4) $-14 + 3 =$

(10) $-5 + 14 =$

(5) $-11 + 11 =$

(11) $-2 + 12 =$

(6) $-1 + 13 =$

(12) $-4 + 10 =$

Section 7. Negative First Term (Addition)

Solve each problem and write the answer in the box.

(1) $-15 + 1 =$

(7) $-11 + 9 =$

(2) $-6 + 5 =$

(8) $-12 + 5 =$

(3) $-15 + 8 =$

(9) $-4 + 14 =$

(4) $-4 + 1 =$

(10) $-1 + 15 =$

(5) $-15 + 2 =$

(11) $-6 + 12 =$

(6) $-4 + 9 =$

(12) $-4 + 15 =$

Section 7. Negative First Term (Addition)

Solve each problem and write the answer in the box.

(1) $-12 + 12 =$

(7) $-12 + 11 =$

(2) $-2 + 13 =$

(8) $-3 + 6 =$

(3) $-1 + 14 =$

(9) $-6 + 6 =$

(4) $-6 + 13 =$

(10) $-2 + 7 =$

(5) $-2 + 3 =$

(11) $-14 + 4 =$

(6) $-9 + 9 =$

(12) $-8 + 2 =$

Section 7. Negative First Term (Addition)

Solve each problem and write the answer in the box.

(1) $-3 + 2 =$

(7) $-13 + 7 =$

(2) $-6 + 5 =$

(8) $-1 + 7 =$

(3) $-12 + 13 =$

(9) $-1 + 4 =$

(4) $-14 + 4 =$

(10) $-12 + 10 =$

(5) $-15 + 3 =$

(11) $-11 + 6 =$

(6) $-7 + 10 =$

(12) $-10 + 5 =$

Section 7. Negative First Term (Addition)

Solve each problem and write the answer in the box.

(1) $-14 + 10 =$

(7) $-14 + 12 =$

(2) $-11 + 13 =$

(8) $-1 + 8 =$

(3) $-8 + 11 =$

(9) $-8 + 5 =$

(4) $-7 + 15 =$

(10) $-2 + 8 =$

(5) $-9 + 5 =$

(11) $-6 + 7 =$

(6) $-6 + 11 =$

(12) $-7 + 7 =$

Section 7. Negative First Term (Addition)

Solve each problem and write the answer in the box.

(1) $-13 + 13 =$

(7) $-5 + 12 =$

(2) $-8 + 13 =$

(8) $-4 + 12 =$

(3) $-14 + 6 =$

(9) $-9 + 11 =$

(4) $-15 + 11 =$

(10) $-11 + 2 =$

(5) $-6 + 2 =$

(11) $-12 + 11 =$

(6) $-5 + 11 =$

(12) $-13 + 10 =$

Section 8. Both Terms Negative (Addition)

Solve each problem and write the answer in the box.

(1) $-2 - 1 =$

(7) $-5 - 11 =$

(2) $-11 - 5 =$

(8) $-15 - 1 =$

(3) $-12 - 12 =$

(9) $-1 - 12 =$

(4) $-10 - 11 =$

(10) $-2 - 11 =$

(5) $-3 - 2 =$

(11) $-13 - 11 =$

(6) $-3 - 4 =$

(12) $-4 - 10 =$

Section 8. Both Terms Negative (Addition)

Solve each problem and write the answer in the box.

(1) $-8 - 4 =$

(7) $-13 - 10 =$

(2) $-4 - 4 =$

(8) $-13 - 14 =$

(3) $-6 - 5 =$

(9) $-5 - 5 =$

(4) $-6 - 9 =$

(10) $-15 - 1 =$

(5) $-1 - 1 =$

(11) $-8 - 13 =$

(6) $-15 - 6 =$

(12) $-7 - 1 =$

Section 8. Both Terms Negative (Addition)

Solve each problem and write the answer in the box.

(1) $-3 - 7 =$

(7) $-13 - 6 =$

(2) $-15 - 5 =$

(8) $-6 - 2 =$

(3) $-1 - 13 =$

(9) $-1 - 3 =$

(4) $-6 - 4 =$

(10) $-3 - 4 =$

(5) $-11 - 7 =$

(11) $-14 - 5 =$

(6) $-14 - 11 =$

(12) $-3 - 13 =$

Section 8. Both Terms Negative (Addition)

Solve each problem and write the answer in the box.

(1) $- 11 - 8 =$

(7) $- 7 - 12 =$

(2) $- 8 - 7 =$

(8) $- 4 - 13 =$

(3) $- 9 - 12 =$

(9) $- 13 - 15 =$

(4) $- 11 - 14 =$

(10) $- 9 - 4 =$

(5) $- 12 - 4 =$

(11) $- 1 - 5 =$

(6) $- 1 - 4 =$

(12) $- 13 - 6 =$

Section 8. Both Terms Negative (Addition)

Solve each problem and write the answer in the box.

(1) $-15 - 4 =$

(7) $-13 - 9 =$

(2) $-9 - 11 =$

(8) $-15 - 11 =$

(3) $-7 - 11 =$

(9) $-8 - 5 =$

(4) $-12 - 13 =$

(10) $-9 - 15 =$

(5) $-3 - 12 =$

(11) $-13 - 12 =$

(6) $-6 - 7 =$

(12) $-13 - 8 =$

Section 8. Both Terms Negative (Addition)

Solve each problem and write the answer in the box.

(1) $-12 - 7 =$

(7) $-10 - 11 =$

(2) $-14 - 11 =$

(8) $-1 - 10 =$

(3) $-6 - 6 =$

(9) $-5 - 4 =$

(4) $-10 - 14 =$

(10) $-15 - 11 =$

(5) $-6 - 8 =$

(11) $-15 - 14 =$

(6) $-14 - 6 =$

(12) $-8 - 5 =$

Section 8. Both Terms Negative (Addition)

Solve each problem and write the answer in the box.

(1) $-6 - 8 =$

(7) $-14 - 15 =$

(2) $-10 - 7 =$

(8) $-2 - 15 =$

(3) $-13 - 2 =$

(9) $-13 - 15 =$

(4) $-14 - 10 =$

(10) $-1 - 15 =$

(5) $-13 - 7 =$

(11) $-8 - 1 =$

(6) $-14 - 14 =$

(12) $-3 - 12 =$

Section 8. Both Terms Negative (Addition)

Solve each problem and write the answer in the box.

(1) $-14 - 13 =$

(7) $-4 - 3 =$

(2) $-12 - 11 =$

(8) $-13 - 3 =$

(3) $-12 - 3 =$

(9) $-4 - 13 =$

(4) $-12 - 14 =$

(10) $-7 - 5 =$

(5) $-4 - 5 =$

(11) $-15 - 15 =$

(6) $-8 - 15 =$

(12) $-15 - 2 =$

Section 8. Both Terms Negative (Addition)

Solve each problem and write the answer in the box.

(1) $-3 - 3 =$

(7) $-9 - 8 =$

(2) $-10 - 9 =$

(8) $-7 - 6 =$

(3) $-12 - 11 =$

(9) $-10 - 4 =$

(4) $-4 - 15 =$

(10) $-2 - 4 =$

(5) $-4 - 1 =$

(11) $-13 - 8 =$

(6) $-2 - 2 =$

(12) $-5 - 11 =$

Section 8. Both Terms Negative (Addition)

Solve each problem and write the answer in the box.

(1) $-13 - 9 =$

(7) $-9 - 2 =$

(2) $-8 - 4 =$

(8) $-12 - 3 =$

(3) $-6 - 11 =$

(9) $-5 - 5 =$

(4) $-5 - 11 =$

(10) $-10 - 2 =$

(5) $-6 - 4 =$

(11) $-12 - 10 =$

(6) $-6 - 12 =$

(12) $-1 - 8 =$

Section 8. Both Terms Negative (Addition)

Solve each problem and write the answer in the box.

(1) $-9 - 13 =$

(7) $-5 - 7 =$

(2) $-15 - 9 =$

(8) $-12 - 10 =$

(3) $-10 - 13 =$

(9) $-4 - 10 =$

(4) $-3 - 11 =$

(10) $-11 - 6 =$

(5) $-4 - 2 =$

(11) $-9 - 10 =$

(6) $-14 - 12 =$

(12) $-14 - 4 =$

Section 8. Both Terms Negative (Addition)

Solve each problem and write the answer in the box.

(1) $-6 - 7 =$

(7) $-11 - 10 =$

(2) $-10 - 4 =$

(8) $-4 - 10 =$

(3) $-10 - 12 =$

(9) $-6 - 1 =$

(4) $-5 - 4 =$

(10) $-2 - 10 =$

(5) $-4 - 9 =$

(11) $-8 - 7 =$

(6) $-11 - 5 =$

(12) $-3 - 11 =$

Section 9. Addition with Brackets (Mixed Signs)

Solve each problem and write the answer in the box.

$$(1) \quad + (-2) - (+3) =$$

$$(7) \quad + (+2) - (-15) =$$

$$(2) \quad + (+15) - (-10) =$$

$$(8) \quad + (-11) - (-9) =$$

$$(3) \quad + (+0) - (+6) =$$

$$(9) \quad + (-6) - (+14) =$$

$$(4) \quad + (+8) - (-6) =$$

$$(10) \quad + (-12) - (+1) =$$

$$(5) \quad + (+3) + (+10) =$$

$$(11) \quad + (-12) + (-11) =$$

$$(6) \quad + (-6) - (+6) =$$

$$(12) \quad + (-9) + (-2) =$$

Section 9. Addition with Brackets (Mixed Signs)

Solve each problem and write the answer in the box.

$$(1) \quad + (+1) - (+0) =$$

$$(7) \quad + (- 8) - (+9) =$$

$$(2) \quad + (- 1) - (+14) =$$

$$(8) \quad + (- 6) - (- 2) =$$

$$(3) \quad + (+4) + (+13) =$$

$$(9) \quad + (+9) + (- 13) =$$

$$(4) \quad + (+15) + (+5) =$$

$$(10) \quad + (+11) + (+1) =$$

$$(5) \quad + (- 6) - (+14) =$$

$$(11) \quad + (- 13) - (+2) =$$

$$(6) \quad + (+8) - (+7) =$$

$$(12) \quad + (- 13) + (- 4) =$$

Section 9. Addition with Brackets (Mixed Signs)

Solve each problem and write the answer in the box.

$$(1) \quad + (+15) + (+12) =$$

$$(7) \quad + (+6) - (+9) =$$

$$(2) \quad + (+7) - (- 4) =$$

$$(8) \quad + (- 9) - (- 2) =$$

$$(3) \quad + (- 1) - (+12) =$$

$$(9) \quad + (- 13) - (+14) =$$

$$(4) \quad + (+15) + (+15) =$$

$$(10) \quad + (+7) - (- 14) =$$

$$(5) \quad + (- 3) + (- 11) =$$

$$(11) \quad + (- 10) - (- 4) =$$

$$(6) \quad + (- 9) - (+13) =$$

$$(12) \quad + (- 13) + (+4) =$$

Section 9. Addition with Brackets (Mixed Signs)

Solve each problem and write the answer in the box.

$$(1) \quad + (+0) - (+15) =$$

$$(7) \quad + (+9) + (- 12) =$$

$$(2) \quad + (+9) - (- 7) =$$

$$(8) \quad + (- 12) - (- 1) =$$

$$(3) \quad + (- 14) - (+15) =$$

$$(9) \quad + (+4) + (- 14) =$$

$$(4) \quad + (- 2) + (- 15) =$$

$$(10) \quad + (+13) - (+10) =$$

$$(5) \quad + (+6) - (+10) =$$

$$(11) \quad + (+8) - (- 6) =$$

$$(6) \quad + (+6) + (+15) =$$

$$(12) \quad + (- 4) + (- 14) =$$

Section 9. Addition with Brackets (Mixed Signs)

Solve each problem and write the answer in the box.

$$(1) \quad + (-13) - (-8) =$$

$$(7) \quad + (-2) + (+11) =$$

$$(2) \quad + (+1) + (+12) =$$

$$(8) \quad + (-5) - (-4) =$$

$$(3) \quad + (+7) - (-13) =$$

$$(9) \quad + (+8) - (-13) =$$

$$(4) \quad + (-6) + (-11) =$$

$$(10) \quad + (+9) + (-15) =$$

$$(5) \quad + (+13) + (+0) =$$

$$(11) \quad + (+1) + (+14) =$$

$$(6) \quad + (+8) + (+1) =$$

$$(12) \quad + (+3) + (-2) =$$

Section 9. Addition with Brackets (Mixed Signs)

Solve each problem and write the answer in the box.

(1) $+ (+3) + (+15) =$

(7) $+ (-3) - (+11) =$

(2) $+ (-11) - (-2) =$

(8) $+ (-10) - (-3) =$

(3) $+ (+4) + (-4) =$

(9) $+ (+7) - (-2) =$

(4) $+ (+4) - (+3) =$

(10) $+ (+14) + (+8) =$

(5) $+ (-1) - (+3) =$

(11) $+ (-4) + (-13) =$

(6) $+ (+1) + (+13) =$

(12) $+ (-12) - (+13) =$

Section 9. Addition with Brackets (Mixed Signs)

Solve each problem and write the answer in the box.

$$(1) \quad + (-3) + (+0) =$$

$$(7) \quad + (+7) + (+5) =$$

$$(2) \quad + (-13) + (-5) =$$

$$(8) \quad + (+15) - (-6) =$$

$$(3) \quad + (-13) - (-15) =$$

$$(9) \quad + (+14) + (-4) =$$

$$(4) \quad + (+4) + (-15) =$$

$$(10) \quad + (-9) - (-2) =$$

$$(5) \quad + (+9) + (-15) =$$

$$(11) \quad + (-2) + (-1) =$$

$$(6) \quad + (-4) - (+11) =$$

$$(12) \quad + (-6) - (-11) =$$

Section 9. Addition with Brackets (Mixed Signs)

Solve each problem and write the answer in the box.

$$(1) \quad + (- 11) + (- 6) =$$

$$(7) \quad + (+10) + (+10) =$$

$$(2) \quad + (- 11) - (+12) =$$

$$(8) \quad + (+10) - (+4) =$$

$$(3) \quad + (- 3) + (+6) =$$

$$(9) \quad + (+2) - (+2) =$$

$$(4) \quad + (- 3) + (- 1) =$$

$$(10) \quad + (- 8) + (+4) =$$

$$(5) \quad + (- 4) - (- 13) =$$

$$(11) \quad + (- 14) - (- 5) =$$

$$(6) \quad + (+2) - (- 7) =$$

$$(12) \quad + (- 8) + (- 10) =$$

Section 9. Addition with Brackets (Mixed Signs)

Solve each problem and write the answer in the box.

$$(1) \quad + (-2) + (-11) =$$

$$(7) \quad + (-10) - (+14) =$$

$$(2) \quad + (-3) + (-14) =$$

$$(8) \quad + (-14) - (-2) =$$

$$(3) \quad + (-14) + (-6) =$$

$$(9) \quad + (+4) - (+15) =$$

$$(4) \quad + (-12) - (+10) =$$

$$(10) \quad + (+2) - (+12) =$$

$$(5) \quad + (-1) + (-7) =$$

$$(11) \quad + (-9) + (+12) =$$

$$(6) \quad + (-4) + (-8) =$$

$$(12) \quad + (-5) + (-11) =$$

Section 9. Addition with Brackets (Mixed Signs)

Solve each problem and write the answer in the box.

$$(1) \quad + (-3) - (-5) =$$

$$(7) \quad + (-12) - (-11) =$$

$$(2) \quad + (+7) + (-2) =$$

$$(8) \quad + (-1) - (-3) =$$

$$(3) \quad + (-14) - (+5) =$$

$$(9) \quad + (-11) + (-14) =$$

$$(4) \quad + (-15) + (+11) =$$

$$(10) \quad + (+11) + (-2) =$$

$$(5) \quad + (-10) - (+11) =$$

$$(11) \quad + (-15) + (+10) =$$

$$(6) \quad + (-15) + (-1) =$$

$$(12) \quad + (+0) - (-14) =$$

Section 9. Addition with Brackets (Mixed Signs)

Solve each problem and write the answer in the box.

$$(1) \quad + (+1) + (- 14) =$$

$$(7) \quad + (+6) + (- 10) =$$

$$(2) \quad + (+12) + (- 1) =$$

$$(8) \quad + (+5) + (- 13) =$$

$$(3) \quad + (- 15) - (+2) =$$

$$(9) \quad + (- 8) + (+15) =$$

$$(4) \quad + (+4) - (- 8) =$$

$$(10) \quad + (- 2) - (+15) =$$

$$(5) \quad + (+8) - (+8) =$$

$$(11) \quad + (- 13) - (+5) =$$

$$(6) \quad + (- 7) - (- 11) =$$

$$(12) \quad + (- 2) + (+3) =$$

Section 9. Addition with Brackets (Mixed Signs)

Solve each problem and write the answer in the box.

$$(1) \quad + (+9) - (- 3) =$$

$$(7) \quad + (+2) - (+4) =$$

$$(2) \quad + (+6) - (- 8) =$$

$$(8) \quad + (+4) + (- 12) =$$

$$(3) \quad + (+15) + (- 1) =$$

$$(9) \quad + (- 9) + (+6) =$$

$$(4) \quad + (- 9) + (+12) =$$

$$(10) \quad + (+6) - (+8) =$$

$$(5) \quad + (+1) - (+9) =$$

$$(11) \quad + (- 8) + (- 9) =$$

$$(6) \quad + (+4) - (+14) =$$

$$(12) \quad + (- 12) - (- 13) =$$

Section 10. Negative Results from Subtraction

Solve each problem and write the answer in the box.

(1) $-10 - 11 =$

(7) $0 - 14 =$

(2) $-10 - 19 =$

(8) $1 - 13 =$

(3) $7 - 19 =$

(9) $9 - 20 =$

(4) $-10 - 13 =$

(10) $-3 - 16 =$

(5) $8 - 16 =$

(11) $3 - 13 =$

(6) $-5 - 20 =$

(12) $2 - 19 =$

Section 10. Negative Results from Subtraction

Solve each problem and write the answer in the box.

(1) $8 - 14 =$

(7) $-4 - 11 =$

(2) $-5 - 18 =$

(8) $-1 - 15 =$

(3) $-5 - 11 =$

(9) $4 - 14 =$

(4) $2 - 20 =$

(10) $-8 - 11 =$

(5) $-7 - 19 =$

(11) $10 - 10 =$

(6) $6 - 19 =$

(12) $7 - 14 =$

Section 10. Negative Results from Subtraction

Solve each problem and write the answer in the box.

(1) $-10 - 14 =$

(7) $1 - 10 =$

(2) $-4 - 17 =$

(8) $-3 - 14 =$

(3) $-3 - 16 =$

(9) $7 - 14 =$

(4) $-4 - 10 =$

(10) $10 - 14 =$

(5) $-2 - 13 =$

(11) $-6 - 15 =$

(6) $9 - 15 =$

(12) $-5 - 19 =$

Section 10. Negative Results from Subtraction

Solve each problem and write the answer in the box.

(1) $9 - 10 =$

(7) $-9 - 13 =$

(2) $-9 - 19 =$

(8) $-6 - 19 =$

(3) $-10 - 18 =$

(9) $-10 - 19 =$

(4) $-10 - 13 =$

(10) $4 - 20 =$

(5) $0 - 15 =$

(11) $5 - 11 =$

(6) $-8 - 16 =$

(12) $4 - 11 =$

Section 10. Negative Results from Subtraction

Solve each problem and write the answer in the box.

(1) $8 - 11 =$

(7) $7 - 13 =$

(2) $1 - 16 =$

(8) $3 - 12 =$

(3) $-7 - 18 =$

(9) $4 - 17 =$

(4) $10 - 17 =$

(10) $8 - 15 =$

(5) $1 - 13 =$

(11) $-9 - 12 =$

(6) $7 - 20 =$

(12) $8 - 17 =$

Section 10. Negative Results from Subtraction

Solve each problem and write the answer in the box.

(1) $-6 - 19 =$

(7) $-2 - 13 =$

(2) $0 - 17 =$

(8) $6 - 18 =$

(3) $9 - 11 =$

(9) $1 - 10 =$

(4) $-1 - 10 =$

(10) $10 - 15 =$

(5) $-10 - 10 =$

(11) $-8 - 14 =$

(6) $0 - 15 =$

(12) $2 - 14 =$

Section 10. Negative Results from Subtraction

Solve each problem and write the answer in the box.

(1) $-9 - 13 =$

(7) $0 - 13 =$

(2) $-2 - 10 =$

(8) $3 - 16 =$

(3) $-10 - 19 =$

(9) $9 - 10 =$

(4) $-8 - 17 =$

(10) $-3 - 15 =$

(5) $3 - 19 =$

(11) $8 - 15 =$

(6) $-4 - 17 =$

(12) $-7 - 17 =$

Section 10. Negative Results from Subtraction

Solve each problem and write the answer in the box.

(1) $1 - 14 =$

(7) $8 - 13 =$

(2) $-6 - 11 =$

(8) $-2 - 19 =$

(3) $6 - 18 =$

(9) $6 - 11 =$

(4) $-8 - 20 =$

(10) $-1 - 14 =$

(5) $-5 - 18 =$

(11) $6 - 15 =$

(6) $9 - 20 =$

(12) $5 - 14 =$

Section 10. Negative Results from Subtraction

Solve each problem and write the answer in the box.

(1) $9 - 15 =$

(7) $-10 - 18 =$

(2) $-1 - 18 =$

(8) $4 - 16 =$

(3) $-2 - 20 =$

(9) $0 - 15 =$

(4) $-4 - 10 =$

(10) $-4 - 16 =$

(5) $5 - 19 =$

(11) $6 - 12 =$

(6) $6 - 10 =$

(12) $1 - 14 =$

Section 10. Negative Results from Subtraction

Solve each problem and write the answer in the box.

(1) $-9 - 15 =$

(7) $3 - 10 =$

(2) $-3 - 12 =$

(8) $8 - 17 =$

(3) $-4 - 15 =$

(9) $1 - 18 =$

(4) $-6 - 12 =$

(10) $-8 - 19 =$

(5) $1 - 20 =$

(11) $2 - 14 =$

(6) $-4 - 16 =$

(12) $-9 - 10 =$

Section 10. Negative Results from Subtraction

Solve each problem and write the answer in the box.

(1) $-6 - 14 =$

(7) $8 - 18 =$

(2) $-1 - 19 =$

(8) $1 - 19 =$

(3) $3 - 18 =$

(9) $-1 - 10 =$

(4) $5 - 10 =$

(10) $-4 - 10 =$

(5) $2 - 16 =$

(11) $-4 - 20 =$

(6) $2 - 18 =$

(12) $-8 - 12 =$

Section 10. Negative Results from Subtraction

Solve each problem and write the answer in the box.

(1) $6 - 10 =$

(7) $6 - 18 =$

(2) $-8 - 15 =$

(8) $-6 - 18 =$

(3) $10 - 19 =$

(9) $3 - 11 =$

(4) $8 - 18 =$

(10) $-7 - 10 =$

(5) $-5 - 12 =$

(11) $-5 - 18 =$

(6) $7 - 14 =$

(12) $-2 - 18 =$

Section 11. Controlled Mixed Signs

Solve each problem and write the answer in the box.

(1) $-1 + 10 =$

(7) $11 - (-10) =$

(2) $-1 - (-11) =$

(8) $3 - 2 =$

(3) $12 - 8 =$

(9) $5 + 1 =$

(4) $-8 - (-9) =$

(10) $2 - (-8) =$

(5) $9 - 1 =$

(11) $-5 - (-2) =$

(6) $11 - (-9) =$

(12) $-2 - 11 =$

Section 11. Controlled Mixed Signs

Solve each problem and write the answer in the box.

(1) $-10 - 3 =$

(7) $8 - (-1) =$

(2) $9 - (-1) =$

(8) $-12 - 7 =$

(3) $-6 - 5 =$

(9) $-4 - 6 =$

(4) $-3 - 10 =$

(10) $-7 - 5 =$

(5) $0 - (-2) =$

(11) $11 + 10 =$

(6) $3 + 12 =$

(12) $-6 + 3 =$

Section 11. Controlled Mixed Signs

Solve each problem and write the answer in the box.

(1) $-1 + 6 =$

(7) $-9 + 11 =$

(2) $0 - 7 =$

(8) $5 - 11 =$

(3) $-8 - 6 =$

(9) $9 - (-5) =$

(4) $0 - (-2) =$

(10) $-2 + 1 =$

(5) $10 - (-1) =$

(11) $6 - 5 =$

(6) $-7 + 7 =$

(12) $-8 - (-4) =$

Section 11. Controlled Mixed Signs

Solve each problem and write the answer in the box.

(1) $-1 - 6 =$

(7) $8 + 7 =$

(2) $12 - (-10) =$

(8) $3 - 7 =$

(3) $0 - 6 =$

(9) $12 - (-9) =$

(4) $0 - 8 =$

(10) $2 - (-11) =$

(5) $-9 - 3 =$

(11) $-11 + 4 =$

(6) $8 - 5 =$

(12) $9 + 4 =$

Section 11. Controlled Mixed Signs

Solve each problem and write the answer in the box.

(1) $7 - 5 =$

(7) $5 + 4 =$

(2) $7 - 6 =$

(8) $7 - 1 =$

(3) $- 3 - 4 =$

(9) $8 - (- 1) =$

(4) $5 + 5 =$

(10) $7 - (- 10) =$

(5) $10 - 10 =$

(11) $8 + 1 =$

(6) $- 2 + 8 =$

(12) $0 - 0 =$

Section 11. Controlled Mixed Signs

Solve each problem and write the answer in the box.

(1) $5 - (-8) =$

(7) $-2 - (-5) =$

(2) $-11 - 7 =$

(8) $-4 - 12 =$

(3) $-4 + 8 =$

(9) $11 - 10 =$

(4) $-4 - 6 =$

(10) $12 - 5 =$

(5) $11 - 12 =$

(11) $2 - 7 =$

(6) $0 - (-1) =$

(12) $0 - (-10) =$

Section 11. Controlled Mixed Signs

Solve each problem and write the answer in the box.

(1) $-12 - (-7) =$

(7) $3 - 6 =$

(2) $-11 - (-5) =$

(8) $2 - 9 =$

(3) $11 - 11 =$

(9) $5 + 6 =$

(4) $-6 - (-11) =$

(10) $2 - (-5) =$

(5) $6 - (-4) =$

(11) $0 - 11 =$

(6) $1 - 5 =$

(12) $-4 - 5 =$

Section 11. Controlled Mixed Signs

Solve each problem and write the answer in the box.

(1) $-5 + 11 =$

(7) $-1 - (-12) =$

(2) $12 - 5 =$

(8) $8 - (-3) =$

(3) $9 - (-4) =$

(9) $-12 + 3 =$

(4) $8 - 12 =$

(10) $5 - 5 =$

(5) $-10 - 8 =$

(11) $-2 + 5 =$

(6) $7 + 2 =$

(12) $-12 + 10 =$

Section 11. Controlled Mixed Signs

Solve each problem and write the answer in the box.

(1) $7 - (-3) =$

(7) $3 - (-4) =$

(2) $5 - 8 =$

(8) $-12 - (-1) =$

(3) $-1 - 7 =$

(9) $8 - 7 =$

(4) $12 - 11 =$

(10) $-3 - 11 =$

(5) $-7 - (-1) =$

(11) $-7 + 6 =$

(6) $1 + 10 =$

(12) $-3 - 8 =$

Section 11. Controlled Mixed Signs

Solve each problem and write the answer in the box.

(1) $- 11 + 3 =$

(7) $- 5 - (- 6) =$

(2) $12 + 0 =$

(8) $12 - 3 =$

(3) $10 - 8 =$

(9) $- 5 - 7 =$

(4) $- 2 - 3 =$

(10) $- 10 - (- 8) =$

(5) $6 - 0 =$

(11) $4 - 10 =$

(6) $- 12 - 5 =$

(12) $6 - 2 =$

Section 11. Controlled Mixed Signs

Solve each problem and write the answer in the box.

(1) $-8 + 2 =$

(7) $-10 - 2 =$

(2) $9 - (-8) =$

(8) $-7 + 12 =$

(3) $-2 + 8 =$

(9) $10 - 3 =$

(4) $5 - 6 =$

(10) $-4 - 0 =$

(5) $4 - (-5) =$

(11) $1 - 6 =$

(6) $7 - 8 =$

(12) $10 - 12 =$

Section 11. Controlled Mixed Signs

Solve each problem and write the answer in the box.

(1) $0 - 10 =$

(7) $- 12 - 11 =$

(2) $4 - 10 =$

(8) $2 - 8 =$

(3) $4 - 7 =$

(9) $9 - (- 3) =$

(4) $3 + 9 =$

(10) $- 1 + 7 =$

(5) $- 2 - 2 =$

(11) $1 - 6 =$

(6) $8 - 5 =$

(12) $12 + 3 =$

Section 12. Mixed with Zero

Solve each problem and write the answer in the box.

(1) $-1 - 12 =$

(7) $-17 - 12 =$

(2) $-20 - 14 =$

(8) $9 - (-14) =$

(3) $9 - (-19) =$

(9) $-17 - 16 =$

(4) $1 - 20 =$

(10) $13 - (-7) =$

(5) $4 - 15 =$

(11) $12 - (-4) =$

(6) $-3 - 1 =$

(12) $14 + 10 =$

Section 12. Mixed with Zero

Solve each problem and write the answer in the box.

(1) $8 - 11 =$

(7) $-12 - 8 =$

(2) $-10 - (-10) =$

(8) $5 - (-9) =$

(3) $-16 - 4 =$

(9) $11 - 20 =$

(4) $-10 - 18 =$

(10) $-4 - 10 =$

(5) $-10 + 11 =$

(11) $13 + 12 =$

(6) $-20 - 14 =$

(12) $-6 - 6 =$

Section 12. Mixed with Zero

Solve each problem and write the answer in the box.

(1) $6 - (-7) =$

(7) $-13 - 16 =$

(2) $9 - (-4) =$

(8) $-10 - 9 =$

(3) $13 - 13 =$

(9) $14 - 19 =$

(4) $0 - 18 =$

(10) $-8 + 4 =$

(5) $-15 - 11 =$

(11) $-2 - 13 =$

(6) $-16 - 8 =$

(12) $11 - 16 =$

Section 12. Mixed with Zero

Solve each problem and write the answer in the box.

(1) $4 - 3 =$

(7) $- 11 - 13 =$

(2) $- 11 + 4 =$

(8) $- 20 - 12 =$

(3) $- 9 + 3 =$

(9) $- 15 - 16 =$

(4) $5 - (- 10) =$

(10) $- 1 - (- 11) =$

(5) $- 7 - 19 =$

(11) $0 - 15 =$

(6) $9 + 13 =$

(12) $19 - 17 =$

Section 12. Mixed with Zero

Solve each problem and write the answer in the box.

(1) $0 - (-19) =$

(7) $13 - 7 =$

(2) $15 + 16 =$

(8) $-11 - 10 =$

(3) $-14 - (-6) =$

(9) $4 + 6 =$

(4) $18 - 16 =$

(10) $-19 - (-18) =$

(5) $-4 - (-17) =$

(11) $-14 - 4 =$

(6) $17 - (-13) =$

(12) $7 - (-20) =$

Section 12. Mixed with Zero

Solve each problem and write the answer in the box.

(1) $20 + 3 =$

(7) $11 - (-6) =$

(2) $-14 - 2 =$

(8) $15 + 3 =$

(3) $12 + 16 =$

(9) $-11 - 12 =$

(4) $9 + 13 =$

(10) $5 - 10 =$

(5) $10 - (-13) =$

(11) $12 - 7 =$

(6) $-7 - (-17) =$

(12) $6 - (-5) =$

Section 12. Mixed with Zero

Solve each problem and write the answer in the box.

(1) $5 + 0 =$

(7) $- 7 - 7 =$

(2) $13 - 10 =$

(8) $5 - (- 16) =$

(3) $20 - 2 =$

(9) $19 - 5 =$

(4) $4 - 19 =$

(10) $10 - (- 9) =$

(5) $- 4 + 0 =$

(11) $11 - 6 =$

(6) $4 + 8 =$

(12) $11 - 1 =$

Section 12. Mixed with Zero

Solve each problem and write the answer in the box.

(1) $5 - 17 =$

(7) $12 - 8 =$

(2) $9 - 18 =$

(8) $20 - 3 =$

(3) $15 - (-6) =$

(9) $6 - 13 =$

(4) $19 - 12 =$

(10) $4 - 5 =$

(5) $-15 - (-8) =$

(11) $13 - (-3) =$

(6) $-6 - 5 =$

(12) $-20 - 2 =$

Section 12. Mixed with Zero

Solve each problem and write the answer in the box.

(1) $12 - (- 14) =$

(7) $- 15 - (- 4) =$

(2) $3 + 18 =$

(8) $2 - 14 =$

(3) $5 - 19 =$

(9) $15 - 10 =$

(4) $4 + 6 =$

(10) $1 - 9 =$

(5) $- 6 + 20 =$

(11) $- 20 - (- 10) =$

(6) $20 - 15 =$

(12) $16 - (- 12) =$

Section 12. Mixed with Zero

Solve each problem and write the answer in the box.

(1) $-18 - 4 =$

(7) $4 - (-7) =$

(2) $-17 - (-14) =$

(8) $-5 - (-12) =$

(3) $-9 - 0 =$

(9) $1 + 6 =$

(4) $-17 - (-1) =$

(10) $13 - 16 =$

(5) $-16 - (-19) =$

(11) $-2 + 10 =$

(6) $1 - (-13) =$

(12) $-4 - (-4) =$

Section 12. Mixed with Zero

Solve each problem and write the answer in the box.

(1) $2 - 3 =$

(7) $6 - (-13) =$

(2) $-20 - (-11) =$

(8) $-9 - 2 =$

(3) $2 - 19 =$

(9) $13 - (-3) =$

(4) $-9 - 7 =$

(10) $5 - (-13) =$

(5) $-7 + 1 =$

(11) $17 - 11 =$

(6) $-2 - 12 =$

(12) $-6 + 1 =$

Section 12. Mixed with Zero

Solve each problem and write the answer in the box.

(1) $-9 - (-11) =$

(7) $6 - 20 =$

(2) $2 - 19 =$

(8) $-9 + 10 =$

(3) $-18 - 1 =$

(9) $-2 - 5 =$

(4) $1 - 13 =$

(10) $-8 + 12 =$

(5) $5 - 10 =$

(11) $9 - (-8) =$

(6) $-7 - (-4) =$

(12) $-13 - 11 =$

Section 13. Mixed Signs: Wider Range

Solve each problem and write the answer in the box.

(1) $-8 - (-24) =$

(7) $27 - 26 =$

(2) $-20 - 25 =$

(8) $-2 - (-10) =$

(3) $-30 - 16 =$

(9) $20 + 5 =$

(4) $20 - 5 =$

(10) $23 - (-11) =$

(5) $-6 - 21 =$

(11) $-19 + 12 =$

(6) $-6 - (-22) =$

(12) $-7 + 14 =$

Section 13. Mixed Signs: Wider Range

Solve each problem and write the answer in the box.

(1) $9 + 9 =$

(7) $-12 - 19 =$

(2) $26 + 18 =$

(8) $-28 + 29 =$

(3) $-27 - (-28) =$

(9) $9 - 11 =$

(4) $-19 + 14 =$

(10) $-23 - 14 =$

(5) $-26 - (-13) =$

(11) $13 - 4 =$

(6) $20 - (-13) =$

(12) $23 - 9 =$

Section 13. Mixed Signs: Wider Range

Solve each problem and write the answer in the box.

(1) $22 - 29 =$

(7) $- 15 + 29 =$

(2) $25 - (- 11) =$

(8) $3 + 30 =$

(3) $8 - 10 =$

(9) $- 12 - 29 =$

(4) $- 19 + 14 =$

(10) $- 19 - 29 =$

(5) $12 - 26 =$

(11) $8 + 3 =$

(6) $- 27 + 22 =$

(12) $15 - 17 =$

Section 13. Mixed Signs: Wider Range

Solve each problem and write the answer in the box.

(1) $-5 + 10 =$

(7) $6 - (-30) =$

(2) $25 - 24 =$

(8) $-14 + 23 =$

(3) $25 + 29 =$

(9) $13 + 27 =$

(4) $20 - (-13) =$

(10) $28 - (-3) =$

(5) $-22 - (-23) =$

(11) $7 - 28 =$

(6) $13 + 3 =$

(12) $-4 + 30 =$

Section 13. Mixed Signs: Wider Range

Solve each problem and write the answer in the box.

(1) $7 + 4 =$

(7) $26 - 16 =$

(2) $-30 - (-4) =$

(8) $12 - (-14) =$

(3) $12 - (-27) =$

(9) $-29 - 18 =$

(4) $-27 - 25 =$

(10) $13 - 18 =$

(5) $-27 - (-28) =$

(11) $26 + 5 =$

(6) $-13 - (-4) =$

(12) $-10 - 6 =$

Section 13. Mixed Signs: Wider Range

Solve each problem and write the answer in the box.

(1) $-19 - 19 =$

(7) $-7 - 8 =$

(2) $-19 - 14 =$

(8) $-20 + 15 =$

(3) $-29 - 18 =$

(9) $24 - 10 =$

(4) $16 - (-5) =$

(10) $-3 - 3 =$

(5) $-18 - (-3) =$

(11) $-14 - (-13) =$

(6) $8 + 0 =$

(12) $7 - 12 =$

Section 13. Mixed Signs: Wider Range

Solve each problem and write the answer in the box.

(1) $-1 - 14 =$

(7) $2 - 29 =$

(2) $-22 - 6 =$

(8) $26 - 17 =$

(3) $22 - 3 =$

(9) $-15 + 7 =$

(4) $-6 - (-23) =$

(10) $18 - 11 =$

(5) $-22 - 0 =$

(11) $-11 - 8 =$

(6) $30 - 8 =$

(12) $14 - (-16) =$

Section 13. Mixed Signs: Wider Range

Solve each problem and write the answer in the box.

(1) $-16 - 6 =$

(7) $-14 - 18 =$

(2) $19 - 28 =$

(8) $-24 - (-30) =$

(3) $22 - 1 =$

(9) $-10 - 23 =$

(4) $-13 - 5 =$

(10) $6 - (-1) =$

(5) $11 - (-12) =$

(11) $-25 - 14 =$

(6) $-6 - (-24) =$

(12) $-19 - (-7) =$

Section 13. Mixed Signs: Wider Range

Solve each problem and write the answer in the box.

(1) $17 - (-16) =$

(7) $-7 - (-28) =$

(2) $-27 - 3 =$

(8) $-16 - 30 =$

(3) $0 - (-21) =$

(9) $22 - 9 =$

(4) $-14 - (-9) =$

(10) $-6 + 30 =$

(5) $-14 - (-25) =$

(11) $-16 - 2 =$

(6) $24 - (-15) =$

(12) $-15 - 7 =$

Section 13. Mixed Signs: Wider Range

Solve each problem and write the answer in the box.

(1) $-27 - 12 =$

(7) $9 - (-30) =$

(2) $-23 - (-18) =$

(8) $0 - (-29) =$

(3) $21 + 0 =$

(9) $-20 + 19 =$

(4) $-28 + 29 =$

(10) $-28 - (-14) =$

(5) $13 + 11 =$

(11) $-18 + 4 =$

(6) $11 - (-6) =$

(12) $-7 - (-5) =$

Section 13. Mixed Signs: Wider Range

Solve each problem and write the answer in the box.

(1) $-19 - (-14) =$

(7) $-2 - 15 =$

(2) $-14 - 8 =$

(8) $4 - 1 =$

(3) $19 - (-7) =$

(9) $-29 - 4 =$

(4) $1 + 6 =$

(10) $25 - (-3) =$

(5) $-16 - 9 =$

(11) $28 - (-1) =$

(6) $-13 - 6 =$

(12) $5 - 30 =$

Section 13. Mixed Signs: Wider Range

Solve each problem and write the answer in the box.

(1) $24 - 5 =$

(7) $13 - 23 =$

(2) $-30 - 4 =$

(8) $22 + 28 =$

(3) $25 - 24 =$

(9) $2 + 19 =$

(4) $-4 - 28 =$

(10) $7 - (-12) =$

(5) $29 + 20 =$

(11) $-13 + 26 =$

(6) $2 - 22 =$

(12) $0 + 29 =$

Section 14. Near Boundaries: -40 to 40

Solve each problem and write the answer in the box.

(1) $-1 - 39 =$

(7) $-40 - (-23) =$

(2) $20 - 30 =$

(8) $6 + 30 =$

(3) $13 - 12 =$

(9) $4 - 1 =$

(4) $-24 - 36 =$

(10) $2 - (-40) =$

(5) $15 + 29 =$

(11) $-9 - (-1) =$

(6) $6 - 27 =$

(12) $35 - 18 =$

Section 14. Near Boundaries: -40 to 40

Solve each problem and write the answer in the box.

(1) $-40 - 2 =$

(7) $-35 - 4 =$

(2) $-28 - (-6) =$

(8) $-9 - (-12) =$

(3) $-8 + 9 =$

(9) $-32 - 40 =$

(4) $-2 + 20 =$

(10) $-39 - 39 =$

(5) $33 - (-24) =$

(11) $-19 + 19 =$

(6) $-16 - 14 =$

(12) $37 - 27 =$

Section 14. Near Boundaries: -40 to 40

Solve each problem and write the answer in the box.

(1) $30 + 36 =$

(7) $- 8 - 30 =$

(2) $4 - (- 23) =$

(8) $- 5 - 31 =$

(3) $- 32 + 25 =$

(9) $31 - 16 =$

(4) $39 - 22 =$

(10) $- 23 - (- 5) =$

(5) $- 30 - (- 16) =$

(11) $30 - 28 =$

(6) $6 - 36 =$

(12) $14 - 20 =$

Section 14. Near Boundaries: -40 to 40

Solve each problem and write the answer in the box.

(1) $40 - 23 =$

(7) $14 - 27 =$

(2) $18 + 23 =$

(8) $2 - 27 =$

(3) $-24 - 28 =$

(9) $-19 - (-19) =$

(4) $11 - (-33) =$

(10) $-3 - (-13) =$

(5) $-1 - 33 =$

(11) $40 - 31 =$

(6) $-31 - (-2) =$

(12) $17 - 21 =$

Section 14. Near Boundaries: -40 to 40

Solve each problem and write the answer in the box.

(1) $35 - 17 =$

(7) $18 - 6 =$

(2) $-33 + 35 =$

(8) $-7 - (-26) =$

(3) $-26 - 0 =$

(9) $-33 + 27 =$

(4) $-16 - 34 =$

(10) $12 - 9 =$

(5) $-28 - 4 =$

(11) $18 - (-1) =$

(6) $-26 - 10 =$

(12) $-40 - 28 =$

Section 14. Near Boundaries: -40 to 40

Solve each problem and write the answer in the box.

(1) $21 - (-11) =$

(7) $-3 - 33 =$

(2) $32 - (-1) =$

(8) $34 - 11 =$

(3) $12 - 5 =$

(9) $37 - 12 =$

(4) $33 + 12 =$

(10) $37 - (-2) =$

(5) $10 - 18 =$

(11) $-18 - 3 =$

(6) $13 - 8 =$

(12) $7 - 15 =$

Section 14. Near Boundaries: -40 to 40

Solve each problem and write the answer in the box.

(1) $-22 - (-18) =$

(7) $30 - 36 =$

(2) $2 - (-2) =$

(8) $-1 - 39 =$

(3) $23 - 16 =$

(9) $-10 - 1 =$

(4) $7 - (-9) =$

(10) $-16 + 38 =$

(5) $-13 - (-18) =$

(11) $40 - 14 =$

(6) $11 + 3 =$

(12) $17 - (-35) =$

Section 14. Near Boundaries: -40 to 40

Solve each problem and write the answer in the box.

(1) $-32 + 20 =$

(7) $-30 - 25 =$

(2) $-4 - 9 =$

(8) $10 - 19 =$

(3) $15 - 29 =$

(9) $-26 - (-29) =$

(4) $30 - 15 =$

(10) $17 - 22 =$

(5) $-20 + 12 =$

(11) $40 - 37 =$

(6) $-38 + 11 =$

(12) $29 - (-21) =$

Section 14. Near Boundaries: -40 to 40

Solve each problem and write the answer in the box.

(1) $-9 - 18 =$

(7) $39 - 21 =$

(2) $-14 + 6 =$

(8) $22 + 12 =$

(3) $-11 - (-35) =$

(9) $-16 - 25 =$

(4) $-17 - 7 =$

(10) $28 + 9 =$

(5) $37 - (-36) =$

(11) $14 - 4 =$

(6) $40 + 35 =$

(12) $37 - (-13) =$

Section 14. Near Boundaries: -40 to 40

Solve each problem and write the answer in the box.

(1) $-15 - 22 =$

(7) $25 - 9 =$

(2) $27 - 8 =$

(8) $-9 - 7 =$

(3) $-38 - (-39) =$

(9) $-18 - (-21) =$

(4) $26 + 21 =$

(10) $8 - 28 =$

(5) $39 - (-36) =$

(11) $32 - (-12) =$

(6) $-35 + 33 =$

(12) $-25 + 29 =$

Section 14. Near Boundaries: -40 to 40

Solve each problem and write the answer in the box.

(1) $29 + 2 =$

(7) $36 - (- 33) =$

(2) $1 + 13 =$

(8) $15 - (- 8) =$

(3) $- 17 - (- 38) =$

(9) $- 40 - 37 =$

(4) $- 16 + 31 =$

(10) $- 11 - 25 =$

(5) $- 35 + 17 =$

(11) $- 11 + 2 =$

(6) $10 - 34 =$

(12) $- 26 + 30 =$

Section 14. Near Boundaries: -40 to 40

Solve each problem and write the answer in the box.

(1) $8 - 8 =$

(7) $36 + 23 =$

(2) $27 - 32 =$

(8) $1 - (-27) =$

(3) $24 - 39 =$

(9) $-17 + 39 =$

(4) $16 - 8 =$

(10) $4 - 16 =$

(5) $13 + 17 =$

(11) $38 - 11 =$

(6) $-10 - 11 =$

(12) $4 - 40 =$

Section 15. Final Revision: All Concepts

Solve each problem and write the answer in the box.

(1) $-23 - 3 =$

(7) $-10 - 29 =$

(2) $2 - (-20) =$

(8) $-11 + 17 =$

(3) $-7 + 5 =$

(9) $-28 - 22 =$

(4) $-12 - 10 =$

(10) $-23 + 29 =$

(5) $3 - (-5) =$

(11) $-22 - (-38) =$

(6) $14 - 7 =$

(12) $14 + 2 =$

Section 15. Final Revision: All Concepts

Solve each problem and write the answer in the box.

(1) $-34 + 7 =$

(7) $12 - 26 =$

(2) $12 - (-39) =$

(8) $3 - (-26) =$

(3) $37 - (-26) =$

(9) $-10 - 7 =$

(4) $20 - 26 =$

(10) $22 - (-10) =$

(5) $29 + 34 =$

(11) $-30 + 24 =$

(6) $23 - (-6) =$

(12) $-4 - 18 =$

Section 15. Final Revision: All Concepts

Solve each problem and write the answer in the box.

(1) $21 - 31 =$

(7) $19 - 11 =$

(2) $17 - (-26) =$

(8) $30 - 6 =$

(3) $24 - (-13) =$

(9) $12 + 11 =$

(4) $6 + 34 =$

(10) $-34 - 25 =$

(5) $-24 + 25 =$

(11) $-37 + 5 =$

(6) $-19 - 35 =$

(12) $26 - (-11) =$

Section 15. Final Revision: All Concepts

Solve each problem and write the answer in the box.

(1) $-2 - (-23) =$

(7) $-15 - (-2) =$

(2) $5 - (-18) =$

(8) $-3 - (-26) =$

(3) $-38 - 30 =$

(9) $-9 - 9 =$

(4) $-25 - 26 =$

(10) $11 - 14 =$

(5) $26 - 12 =$

(11) $8 - (-38) =$

(6) $-30 - 25 =$

(12) $-17 - 37 =$

Section 15. Final Revision: All Concepts

Solve each problem and write the answer in the box.

(1) $-17 - (-18) =$

(7) $38 - (-38) =$

(2) $-27 - (-3) =$

(8) $24 - (-20) =$

(3) $-19 + 5 =$

(9) $5 - 6 =$

(4) $-27 - 11 =$

(10) $-35 - 31 =$

(5) $-30 - 6 =$

(11) $-29 - (-11) =$

(6) $29 - 39 =$

(12) $13 + 33 =$

Section 15. Final Revision: All Concepts

Solve each problem and write the answer in the box.

(1) $1 - (-24) =$

(7) $-30 + 2 =$

(2) $-14 + 13 =$

(8) $-37 - 28 =$

(3) $13 + 1 =$

(9) $-11 + 7 =$

(4) $18 - 17 =$

(10) $-22 + 34 =$

(5) $0 + 0 =$

(11) $37 - 8 =$

(6) $22 + 26 =$

(12) $-40 - 4 =$

Section 15. Final Revision: All Concepts

Solve each problem and write the answer in the box.

(1) $-12 - (-19) =$

(7) $-17 - 5 =$

(2) $32 + 17 =$

(8) $-12 - 27 =$

(3) $-13 - 29 =$

(9) $-33 - 30 =$

(4) $-18 + 20 =$

(10) $-20 - (-2) =$

(5) $35 + 33 =$

(11) $31 + 31 =$

(6) $18 - 6 =$

(12) $7 + 0 =$

Section 15. Final Revision: All Concepts

Solve each problem and write the answer in the box.

(1) $21 - 0 =$

(7) $5 - 33 =$

(2) $-9 - (-25) =$

(8) $8 - 17 =$

(3) $-9 - 6 =$

(9) $7 - (-20) =$

(4) $35 + 39 =$

(10) $-23 - 31 =$

(5) $-35 - (-10) =$

(11) $-21 + 9 =$

(6) $12 - 19 =$

(12) $-14 + 37 =$

Section 15. Final Revision: All Concepts

Solve each problem and write the answer in the box.

(1) $-14 - 7 =$

(7) $2 + 31 =$

(2) $11 - (-33) =$

(8) $9 + 36 =$

(3) $-39 - 33 =$

(9) $32 + 1 =$

(4) $-6 + 30 =$

(10) $29 + 16 =$

(5) $-2 - 14 =$

(11) $-32 - 1 =$

(6) $-23 + 29 =$

(12) $35 - (-20) =$

Section 15. Final Revision: All Concepts

Solve each problem and write the answer in the box.

(1) $-37 - 16 =$

(7) $40 + 18 =$

(2) $-1 - (-27) =$

(8) $-9 - 3 =$

(3) $19 - (-1) =$

(9) $16 - 1 =$

(4) $4 - 3 =$

(10) $-7 + 6 =$

(5) $-20 - 27 =$

(11) $20 - 7 =$

(6) $38 - 37 =$

(12) $-6 - 35 =$

Section 15. Final Revision: All Concepts

Solve each problem and write the answer in the box.

(1) $18 - 2 =$

(7) $- 2 + 22 =$

(2) $- 25 - (- 7) =$

(8) $4 - (- 16) =$

(3) $- 29 - (- 36) =$

(9) $- 20 - 40 =$

(4) $0 - (- 5) =$

(10) $36 - (- 19) =$

(5) $- 30 - (- 36) =$

(11) $3 - 29 =$

(6) $- 8 - (- 21) =$

(12) $- 23 - 25 =$

Section 15. Final Revision: All Concepts

Solve each problem and write the answer in the box.

(1) $26 - 35 =$

(7) $26 + 10 =$

(2) $6 + 2 =$

(8) $- 10 - 7 =$

(3) $- 32 - (- 8) =$

(9) $31 + 40 =$

(4) $- 1 - 12 =$

(10) $26 - 1 =$

(5) $29 + 4 =$

(11) $- 1 - 10 =$

(6) $37 + 7 =$

(12) $- 16 - 7 =$

1a-1	1	2a-1	-2	3a-1	0	4a-1	0
1a-2	0	2a-2	3	3a-2	0	4a-2	7
1a-3	0	2a-3	-1	3a-3	-1	4a-3	0
1a-4	0	2a-4	1	3a-4	1	4a-4	0
1a-5	0	2a-5	5	3a-5	6	4a-5	0
1a-6	0	2a-6	4	3a-6	2	4a-6	6
1a-7	2	2a-7	2	3a-7	1	4a-7	0
1a-8	0	2a-8	-2	3a-8	3	4a-8	8
1a-9	-2	2a-9	6	3a-9	3	4a-9	4
1a-10	-2	2a-10	-1	3a-10	-2	4a-10	4
1a-11	2	2a-11	1	3a-11	3	4a-11	6
1a-12	3	2a-12	4	3a-12	2	4a-12	4
1b-1	3	2b-1	0	3b-1	-2	4b-1	-2
1b-2	-2	2b-2	-1	3b-2	0	4b-2	-2
1b-3	5	2b-3	2	3b-3	-2	4b-3	-2
1b-4	2	2b-4	-1	3b-4	0	4b-4	0
1b-5	3	2b-5	0	3b-5	3	4b-5	1
1b-6	2	2b-6	1	3b-6	0	4b-6	0
1b-7	1	2b-7	-2	3b-7	4	4b-7	6
1b-8	2	2b-8	-1	3b-8	1	4b-8	-1
1b-9	0	2b-9	-2	3b-9	7	4b-9	6
1b-10	0	2b-10	-1	3b-10	1	4b-10	-2
1b-11	5	2b-11	-2	3b-11	1	4b-11	0
1b-12	-1	2b-12	0	3b-12	2	4b-12	-1

5a-1	0	6a-1	0	7a-1	-6	8a-1	-5
5a-2	0	6a-2	-1	7a-2	0	8a-2	13
5a-3	-2	6a-3	-1	7a-3	10	8a-3	8
5a-4	0	6a-4	3	7a-4	-5	8a-4	4
5a-5	-2	6a-5	0	7a-5	-1	8a-5	1
5a-6	2	6a-6	4	7a-6	-9	8a-6	9
5a-7	3	6a-7	6	7a-7	-3	8a-7	1
5a-8	0	6a-8	0	7a-8	1	8a-8	-5
5a-9	0	6a-9	-1	7a-9	-1	8a-9	-10
5a-10	0	6a-10	1	7a-10	-11	8a-10	-8
5a-11	1	6a-11	0	7a-11	-4	8a-11	3
5a-12	8	6a-12	3	7a-12	2	8a-12	0
5b-1	-1	6b-1	0	7b-1	-4	8b-1	1
5b-2	1	6b-2	8	7b-2	-6	8b-2	0
5b-3	1	6b-3	2	7b-3	11	8b-3	6
5b-4	-1	6b-4	3	7b-4	3	8b-4	2
5b-5	5	6b-5	4	7b-5	3	8b-5	0
5b-6	0	6b-6	5	7b-6	-1	8b-6	13
5b-7	-2	6b-7	3	7b-7	-8	8b-7	-3
5b-8	6	6b-8	1	7b-8	-7	8b-8	-6
5b-9	3	6b-9	-1	7b-9	-1	8b-9	-1
5b-10	4	6b-10	4	7b-10	-1	8b-10	7
5b-11	6	6b-11	3	7b-11	5	8b-11	-5
5b-12	-1	6b-12	-1	7b-12	-2	8b-12	10

9a-1	4	10a-1	3	11a-1	-5	12a-1	10
9a-2	-3	10a-2	9	11a-2	5	12a-2	4
9a-3	3	10a-3	6	11a-3	-5	12a-3	0
9a-4	-3	10a-4	-5	11a-4	-7	12a-4	1
9a-5	9	10a-5	11	11a-5	3	12a-5	7
9a-6	-5	10a-6	0	11a-6	1	12a-6	-5
9a-7	7	10a-7	6	11a-7	0	12a-7	11
9a-8	1	10a-8	-3	11a-8	9	12a-8	6
9a-9	6	10a-9	-8	11a-9	0	12a-9	-3
9a-10	7	10a-10	-8	11a-10	8	12a-10	7
9a-11	0	10a-11	-12	11a-11	0	12a-11	-8
9a-12	0	10a-12	3	11a-12	-13	12a-12	-1
9b-1	-11	10b-1	12	11b-1	-2	12b-1	12
9b-2	0	10b-2	6	11b-2	5	12b-2	-5
9b-3	11	10b-3	-8	11b-3	2	12b-3	10
9b-4	4	10b-4	0	11b-4	-3	12b-4	-4
9b-5	1	10b-5	-2	11b-5	0	12b-5	-12
9b-6	5	10b-6	3	11b-6	9	12b-6	-7
9b-7	-2	10b-7	0	11b-7	1	12b-7	6
9b-8	-2	10b-8	-9	11b-8	-5	12b-8	-4
9b-9	-2	10b-9	3	11b-9	-9	12b-9	11
9b-10	9	10b-10	-3	11b-10	4	12b-10	9
9b-11	7	10b-11	-13	11b-11	7	12b-11	-7
9b-12	-5	10b-12	-1	11b-12	-9	12b-12	2

13a-1	0	14a-1	-1	15a-1	0	16a-1	7
13a-2	-2	14a-2	4	15a-2	-5	16a-2	2
13a-3	-1	14a-3	2	15a-3	6	16a-3	0
13a-4	2	14a-4	1	15a-4	-1	16a-4	3
13a-5	2	14a-5	1	15a-5	-4	16a-5	-4
13a-6	0	14a-6	-4	15a-6	4	16a-6	2
13a-7	-2	14a-7	2	15a-7	6	16a-7	6
13a-8	-1	14a-8	-1	15a-8	2	16a-8	-5
13a-9	-4	14a-9	3	15a-9	3	16a-9	0
13a-10	0	14a-10	-2	15a-10	1	16a-10	-6
13a-11	-5	14a-11	2	15a-11	5	16a-11	1
13a-12	1	14a-12	7	15a-12	-3	16a-12	-5
13b-1	6	14b-1	5	15b-1	2	16b-1	0
13b-2	3	14b-2	0	15b-2	6	16b-2	-2
13b-3	4	14b-3	-4	15b-3	0	16b-3	3
13b-4	-4	14b-4	-2	15b-4	-2	16b-4	-2
13b-5	0	14b-5	1	15b-5	2	16b-5	-3
13b-6	1	14b-6	0	15b-6	1	16b-6	1
13b-7	1	14b-7	4	15b-7	2	16b-7	-3
13b-8	3	14b-8	-6	15b-8	-4	16b-8	-1
13b-9	-3	14b-9	0	15b-9	-2	16b-9	-2
13b-10	2	14b-10	5	15b-10	-7	16b-10	1
13b-11	-1	14b-11	-5	15b-11	-3	16b-11	4
13b-12	0	14b-12	-3	15b-12	-1	16b-12	-1

17a-1	4	18a-1	-3	19a-1	-4	20a-1	-3
17a-2	3	18a-2	0	19a-2	3	20a-2	-2
17a-3	-4	18a-3	0	19a-3	2	20a-3	6
17a-4	-6	18a-4	-2	19a-4	-8	20a-4	-7
17a-5	-2	18a-5	3	19a-5	4	20a-5	8
17a-6	6	18a-6	-2	19a-6	5	20a-6	-1
17a-7	-3	18a-7	-6	19a-7	-1	20a-7	8
17a-8	1	18a-8	-3	19a-8	0	20a-8	-5
17a-9	7	18a-9	2	19a-9	-3	20a-9	3
17a-10	3	18a-10	-1	19a-10	3	20a-10	1
17a-11	3	18a-11	7	19a-11	5	20a-11	-1
17a-12	-2	18a-12	1	19a-12	-3	20a-12	5
17b-1	4	18b-1	2	19b-1	-5	20b-1	0
17b-2	-2	18b-2	-4	19b-2	-4	20b-2	-4
17b-3	2	18b-3	-1	19b-3	-9	20b-3	5
17b-4	-2	18b-4	3	19b-4	0	20b-4	9
17b-5	3	18b-5	-1	19b-5	4	20b-5	-1
17b-6	-2	18b-6	-2	19b-6	-1	20b-6	1
17b-7	-3	18b-7	1	19b-7	7	20b-7	-2
17b-8	1	18b-8	-3	19b-8	0	20b-8	-1
17b-9	-4	18b-9	4	19b-9	-1	20b-9	5
17b-10	1	18b-10	2	19b-10	3	20b-10	0
17b-11	0	18b-11	-4	19b-11	-2	20b-11	-6
17b-12	0	18b-12	0	19b-12	-8	20b-12	-2

21a-1	0	22a-1	0	23a-1	2	24a-1	1
21a-2	-3	22a-2	2	23a-2	6	24a-2	3
21a-3	3	22a-3	-3	23a-3	2	24a-3	1
21a-4	3	22a-4	1	23a-4	-5	24a-4	-3
21a-5	4	22a-5	6	23a-5	-7	24a-5	2
21a-6	5	22a-6	8	23a-6	1	24a-6	-2
21a-7	6	22a-7	-3	23a-7	7	24a-7	-5
21a-8	-6	22a-8	-8	23a-8	-7	24a-8	1
21a-9	2	22a-9	-2	23a-9	0	24a-9	-2
21a-10	7	22a-10	-5	23a-10	-5	24a-10	0
21a-11	1	22a-11	-1	23a-11	-1	24a-11	9
21a-12	4	22a-12	-1	23a-12	6	24a-12	2
21b-1	-4	22b-1	7	23b-1	-5	24b-1	-1
21b-2	-5	22b-2	0	23b-2	-3	24b-2	2
21b-3	0	22b-3	2	23b-3	-9	24b-3	5
21b-4	3	22b-4	-3	23b-4	-2	24b-4	-1
21b-5	-2	22b-5	1	23b-5	3	24b-5	1
21b-6	1	22b-6	1	23b-6	-2	24b-6	2
21b-7	1	22b-7	-3	23b-7	2	24b-7	4
21b-8	0	22b-8	-1	23b-8	8	24b-8	2
21b-9	3	22b-9	-3	23b-9	-4	24b-9	-6
21b-10	2	22b-10	0	23b-10	-1	24b-10	5
21b-11	-2	22b-11	-3	23b-11	-1	24b-11	-4
21b-12	-2	22b-12	0	23b-12	-2	24b-12	4

25a-1	9	26a-1	-2	27a-1	4	28a-1	-3
25a-2	9	26a-2	6	27a-2	11	28a-2	4
25a-3	4	26a-3	5	27a-3	10	28a-3	11
25a-4	10	26a-4	1	27a-4	6	28a-4	-1
25a-5	2	26a-5	1	27a-5	8	28a-5	0
25a-6	8	26a-6	14	27a-6	12	28a-6	2
25a-7	-1	26a-7	10	27a-7	13	28a-7	11
25a-8	-2	26a-8	6	27a-8	5	28a-8	3
25a-9	0	26a-9	1	27a-9	2	28a-9	-3
25a-10	-1	26a-10	10	27a-10	9	28a-10	9
25a-11	-4	26a-11	1	27a-11	4	28a-11	10
25a-12	-4	26a-12	5	27a-12	-2	28a-12	7
25b-1	1	26b-1	15	27b-1	2	28b-1	3
25b-2	6	26b-2	10	27b-2	-3	28b-2	-1
25b-3	9	26b-3	8	27b-3	8	28b-3	8
25b-4	-4	26b-4	17	27b-4	10	28b-4	11
25b-5	13	26b-5	12	27b-5	13	28b-5	8
25b-6	0	26b-6	-3	27b-6	13	28b-6	5
25b-7	11	26b-7	4	27b-7	11	28b-7	18
25b-8	-5	26b-8	18	27b-8	-2	28b-8	8
25b-9	-8	26b-9	10	27b-9	17	28b-9	-2
25b-10	5	26b-10	4	27b-10	-6	28b-10	0
25b-11	14	26b-11	-1	27b-11	5	28b-11	15
25b-12	11	26b-12	7	27b-12	4	28b-12	-1

29a-1	-3	30a-1	6	31a-1	0	32a-1	7
29a-2	9	30a-2	13	31a-2	2	32a-2	-1
29a-3	8	30a-3	7	31a-3	7	32a-3	6
29a-4	10	30a-4	4	31a-4	0	32a-4	-7
29a-5	14	30a-5	-2	31a-5	-7	32a-5	-4
29a-6	6	30a-6	15	31a-6	-8	32a-6	-10
29a-7	15	30a-7	0	31a-7	-1	32a-7	1
29a-8	-5	30a-8	16	31a-8	-3	32a-8	-11
29a-9	11	30a-9	-3	31a-9	-2	32a-9	-6
29a-10	12	30a-10	2	31a-10	-5	32a-10	-11
29a-11	8	30a-11	2	31a-11	-11	32a-11	4
29a-12	14	30a-12	5	31a-12	-9	32a-12	1
29b-1	10	30b-1	17	31b-1	-6	32b-1	-10
29b-2	14	30b-2	13	31b-2	2	32b-2	0
29b-3	7	30b-3	15	31b-3	1	32b-3	1
29b-4	5	30b-4	-3	31b-4	-1	32b-4	-2
29b-5	10	30b-5	2	31b-5	2	32b-5	-11
29b-6	5	30b-6	5	31b-6	-10	32b-6	8
29b-7	3	30b-7	1	31b-7	-9	32b-7	-6
29b-8	11	30b-8	-2	31b-8	-10	32b-8	-6
29b-9	17	30b-9	12	31b-9	-3	32b-9	-9
29b-10	9	30b-10	14	31b-10	1	32b-10	-7
29b-11	-6	30b-11	3	31b-11	-12	32b-11	2
29b-12	12	30b-12	2	31b-12	4	32b-12	-1

33a-1	-3	34a-1	-11	35a-1	-2	36a-1	-2
33a-2	-1	34a-2	0	35a-2	-8	36a-2	0
33a-3	-6	34a-3	-1	35a-3	-2	36a-3	-5
33a-4	-9	34a-4	3	35a-4	-9	36a-4	2
33a-5	3	34a-5	-5	35a-5	-10	36a-5	-12
33a-6	5	34a-6	5	35a-6	-4	36a-6	3
33a-7	-8	34a-7	-7	35a-7	2	36a-7	-3
33a-8	-5	34a-8	-4	35a-8	-3	36a-8	-6
33a-9	-1	34a-9	2	35a-9	-6	36a-9	-1
33a-10	-7	34a-10	6	35a-10	0	36a-10	-5
33a-11	-1	34a-11	-4	35a-11	6	36a-11	0
33a-12	-8	34a-12	-6	35a-12	-2	36a-12	1
33b-1	-12	34b-1	-7	35b-1	-10	36b-1	8
33b-2	-8	34b-2	-13	35b-2	-6	36b-2	-10
33b-3	-4	34b-3	7	35b-3	-7	36b-3	-4
33b-4	5	34b-4	2	35b-4	-4	36b-4	-4
33b-5	-9	34b-5	2	35b-5	1	36b-5	-1
33b-6	-1	34b-6	1	35b-6	-6	36b-6	-1
33b-7	-13	34b-7	3	35b-7	1	36b-7	6
33b-8	-1	34b-8	-4	35b-8	6	36b-8	-6
33b-9	6	34b-9	6	35b-9	2	36b-9	-9
33b-10	8	34b-10	1	35b-10	2	36b-10	3
33b-11	-2	34b-11	-8	35b-11	-7	36b-11	0
33b-12	-4	34b-12	0	35b-12	-5	36b-12	-8

37a-1	3	38a-1	-1	39a-1	-9	40a-1	7
37a-2	-14	38a-2	4	39a-2	-4	40a-2	-1
37a-3	-5	38a-3	7	39a-3	0	40a-3	-1
37a-4	-8	38a-4	-5	39a-4	-7	40a-4	-11
37a-5	2	38a-5	-1	39a-5	2	40a-5	0
37a-6	6	38a-6	1	39a-6	1	40a-6	12
37a-7	-8	38a-7	-7	39a-7	-6	40a-7	7
37a-8	10	38a-8	-13	39a-8	7	40a-8	-7
37a-9	-8	38a-9	2	39a-9	10	40a-9	-2
37a-10	-4	38a-10	-8	39a-10	3	40a-10	9
37a-11	-5	38a-11	-2	39a-11	-7	40a-11	10
37a-12	9	38a-12	0	39a-12	0	40a-12	6
37b-1	-2	38b-1	-3	39b-1	5	40b-1	-14
37b-2	-3	38b-2	4	39b-2	3	40b-2	-1
37b-3	-8	38b-3	-4	39b-3	3	40b-3	-7
37b-4	-4	38b-4	11	39b-4	-1	40b-4	-3
37b-5	2	38b-5	-11	39b-5	0	40b-5	-13
37b-6	9	38b-6	-2	39b-6	-11	40b-6	5
37b-7	13	38b-7	2	39b-7	2	40b-7	-2
37b-8	6	38b-8	11	39b-8	5	40b-8	-7
37b-9	-7	38b-9	3	39b-9	1	40b-9	10
37b-10	-3	38b-10	1	39b-10	-5	40b-10	14
37b-11	-5	38b-11	-5	39b-11	-1	40b-11	6
37b-12	-9	38b-12	-3	39b-12	-3	40b-12	11

41a-1	0	42a-1	-4	43a-1	-3	44a-1	-10
41a-2	11	42a-2	2	43a-2	-16	44a-2	-20
41a-3	13	42a-3	3	43a-3	-24	44a-3	-14
41a-4	7	42a-4	8	43a-4	-21	44a-4	-10
41a-5	1	42a-5	-4	43a-5	-5	44a-5	-18
41a-6	0	42a-6	5	43a-6	-7	44a-6	-25
41a-7	-1	42a-7	-2	43a-7	-16	44a-7	-19
41a-8	3	42a-8	7	43a-8	-16	44a-8	-8
41a-9	0	42a-9	-3	43a-9	-13	44a-9	-4
41a-10	5	42a-10	6	43a-10	-13	44a-10	-7
41a-11	-10	42a-11	1	43a-11	-24	44a-11	-19
41a-12	-6	42a-12	0	43a-12	-14	44a-12	-16
41b-1	-1	42b-1	0	43b-1	-12	44b-1	-19
41b-2	-1	42b-2	5	43b-2	-8	44b-2	-15
41b-3	1	42b-3	-8	43b-3	-11	44b-3	-21
41b-4	-10	42b-4	-4	43b-4	-15	44b-4	-25
41b-5	-12	42b-5	-4	43b-5	-2	44b-5	-16
41b-6	3	42b-6	6	43b-6	-21	44b-6	-5
41b-7	-6	42b-7	7	43b-7	-23	44b-7	-19
41b-8	6	42b-8	8	43b-8	-27	44b-8	-17
41b-9	3	42b-9	2	43b-9	-10	44b-9	-28
41b-10	-2	42b-10	-9	43b-10	-16	44b-10	-13
41b-11	-5	42b-11	-1	43b-11	-21	44b-11	-6
41b-12	-5	42b-12	-3	43b-12	-8	44b-12	-19

45a-1	-19	46a-1	-14	47a-1	-6	48a-1	-22
45a-2	-20	46a-2	-17	47a-2	-19	48a-2	-24
45a-3	-18	46a-3	-15	47a-3	-23	48a-3	-23
45a-4	-25	46a-4	-24	47a-4	-19	48a-4	-14
45a-5	-15	46a-5	-20	47a-5	-5	48a-5	-6
45a-6	-13	46a-6	-28	47a-6	-4	48a-6	-26
45a-7	-22	46a-7	-29	47a-7	-17	48a-7	-12
45a-8	-26	46a-8	-17	47a-8	-13	48a-8	-22
45a-9	-13	46a-9	-28	47a-9	-14	48a-9	-14
45a-10	-24	46a-10	-16	47a-10	-6	48a-10	-17
45a-11	-25	46a-11	-9	47a-11	-21	48a-11	-19
45a-12	-21	46a-12	-15	47a-12	-16	48a-12	-18
45b-1	-19	46b-1	-27	47b-1	-22	48b-1	-13
45b-2	-25	46b-2	-23	47b-2	-12	48b-2	-14
45b-3	-12	46b-3	-15	47b-3	-17	48b-3	-22
45b-4	-24	46b-4	-26	47b-4	-16	48b-4	-9
45b-5	-14	46b-5	-9	47b-5	-10	48b-5	-13
45b-6	-20	46b-6	-23	47b-6	-18	48b-6	-16
45b-7	-21	46b-7	-7	47b-7	-11	48b-7	-21
45b-8	-11	46b-8	-16	47b-8	-15	48b-8	-14
45b-9	-9	46b-9	-17	47b-9	-10	48b-9	-7
45b-10	-26	46b-10	-12	47b-10	-12	48b-10	-12
45b-11	-29	46b-11	-30	47b-11	-22	48b-11	-15
45b-12	-13	46b-12	-17	47b-12	-9	48b-12	-14

49a-1	-5	50a-1	27	51a-1	-5	52a-1	-3
49a-2	25	50a-2	11	51a-2	13	52a-2	-18
49a-3	-6	50a-3	-13	51a-3	20	52a-3	2
49a-4	14	50a-4	30	51a-4	-17	52a-4	-11
49a-5	13	50a-5	-14	51a-5	13	52a-5	-6
49a-6	-12	50a-6	-22	51a-6	9	52a-6	-15
49a-7	17	50a-7	-3	51a-7	9	52a-7	12
49a-8	-2	50a-8	-7	51a-8	-1	52a-8	21
49a-9	-20	50a-9	-27	51a-9	21	52a-9	10
49a-10	-13	50a-10	21	51a-10	-6	52a-10	-7
49a-11	-23	50a-11	-6	51a-11	15	52a-11	-3
49a-12	-11	50a-12	-9	51a-12	1	52a-12	5
49b-1	1	50b-1	-15	51b-1	18	52b-1	-17
49b-2	-15	50b-2	16	51b-2	-9	52b-2	-23
49b-3	17	50b-3	-29	51b-3	0	52b-3	3
49b-4	20	50b-4	-17	51b-4	1	52b-4	-4
49b-5	-20	50b-5	-4	51b-5	-4	52b-5	9
49b-6	1	50b-6	21	51b-6	14	52b-6	9
49b-7	-17	50b-7	-3	51b-7	-14	52b-7	20
49b-8	-4	50b-8	-11	51b-8	-7	52b-8	6
49b-9	-4	50b-9	-10	51b-9	9	52b-9	0
49b-10	12	50b-10	3	51b-10	22	52b-10	-4
49b-11	-15	50b-11	14	51b-11	-17	52b-11	-9
49b-12	-17	50b-12	-18	51b-12	-25	52b-12	-18

53a-1	-13	54a-1	-13	55a-1	-21	56a-1	-24
53a-2	-17	54a-2	11	55a-2	-29	56a-2	-21
53a-3	-20	54a-3	-17	55a-3	-12	56a-3	-19
53a-4	-22	54a-4	12	55a-4	-23	56a-4	-14
53a-5	-8	54a-5	0	55a-5	-8	56a-5	-15
53a-6	-12	54a-6	4	55a-6	-25	56a-6	-6
53a-7	-24	54a-7	-4	55a-7	-14	56a-7	-9
53a-8	-12	54a-8	-8	55a-8	-12	56a-8	-17
53a-9	-11	54a-9	7	55a-9	-11	56a-9	-7
53a-10	-10	54a-10	-17	55a-10	-19	56a-10	-4
53a-11	3	54a-11	-18	55a-11	-10	56a-11	-21
53a-12	-16	54a-12	1	55a-12	-17	56a-12	-24
53b-1	2	54b-1	12	55b-1	-6	56b-1	-1
53b-2	5	54b-2	14	55b-2	-23	56b-2	-28
53b-3	-19	54b-3	14	55b-3	-16	56b-3	-28
53b-4	-4	54b-4	3	55b-4	-18	56b-4	-23
53b-5	-21	54b-5	-8	55b-5	-26	56b-5	-15
53b-6	-16	54b-6	-10	55b-6	-13	56b-6	-24
53b-7	-1	54b-7	-2	55b-7	-15	56b-7	-22
53b-8	2	54b-8	-8	55b-8	-16	56b-8	-25
53b-9	-25	54b-9	-3	55b-9	-10	56b-9	-29
53b-10	9	54b-10	-2	55b-10	-19	56b-10	-16
53b-11	-5	54b-11	-17	55b-11	0	56b-11	-6
53b-12	14	54b-12	1	55b-12	-7	56b-12	-7

57a-1	-3	58a-1	-22	59a-1	-6	60a-1	-20
57a-2	-15	58a-2	-12	59a-2	-19	60a-2	-20
57a-3	-25	58a-3	-29	59a-3	-22	60a-3	-15
57a-4	-7	58a-4	-25	59a-4	-14	60a-4	-5
57a-5	-12	58a-5	-16	59a-5	-14	60a-5	-14
57a-6	-13	58a-6	-21	59a-6	-4	60a-6	-16
57a-7	-6	58a-7	-13	59a-7	-28	60a-7	-10
57a-8	-9	58a-8	-13	59a-8	-12	60a-8	-18
57a-9	-13	58a-9	-1	59a-9	-15	60a-9	-11
57a-10	-7	58a-10	-18	59a-10	-20	60a-10	-14
57a-11	-21	58a-11	-7	59a-11	-6	60a-11	-24
57a-12	-9	58a-12	-24	59a-12	-13	60a-12	-20
57b-1	-25	58b-1	-13	59b-1	-24	60b-1	-4
57b-2	-17	58b-2	-17	59b-2	-15	60b-2	-23
57b-3	-2	58b-3	-12	59b-3	-19	60b-3	-9
57b-4	-11	58b-4	-28	59b-4	-18	60b-4	-10
57b-5	-20	58b-5	-23	59b-5	-19	60b-5	-17
57b-6	-15	58b-6	-11	59b-6	-20	60b-6	-7
57b-7	-15	58b-7	-5	59b-7	-7	60b-7	-12
57b-8	-12	58b-8	-21	59b-8	-9	60b-8	-24
57b-9	-9	58b-9	-5	59b-9	-17	60b-9	-8
57b-10	-5	58b-10	-15	59b-10	-27	60b-10	-17
57b-11	-22	58b-11	-9	59b-11	-12	60b-11	-23
57b-12	-12	58b-12	-9	59b-12	-19	60b-12	-20

61a-1	9	62a-1	5	63a-1	2	64a-1	-5
61a-2	10	62a-2	-7	63a-2	1	64a-2	-6
61a-3	4	62a-3	-14	63a-3	-7	64a-3	0
61a-4	1	62a-4	2	63a-4	10	64a-4	5
61a-5	8	62a-5	11	63a-5	0	64a-5	10
61a-6	20	62a-6	0	63a-6	6	64a-6	-4
61a-7	21	62a-7	2	63a-7	9	64a-7	-3
61a-8	1	62a-8	-6	63a-8	6	64a-8	-7
61a-9	6	62a-9	14	63a-9	9	64a-9	11
61a-10	10	62a-10	-1	63a-10	17	64a-10	7
61a-11	-3	62a-11	1	63a-11	9	64a-11	-11
61a-12	-13	62a-12	-4	63a-12	0	64a-12	-9
61b-1	-13	62b-1	-7	63b-1	13	64b-1	6
61b-2	10	62b-2	22	63b-2	-18	64b-2	7
61b-3	-11	62b-3	-6	63b-3	4	64b-3	13
61b-4	-13	62b-4	-8	63b-4	-10	64b-4	-4
61b-5	2	62b-5	-12	63b-5	-1	64b-5	-18
61b-6	15	62b-6	3	63b-6	1	64b-6	9
61b-7	9	62b-7	15	63b-7	3	64b-7	11
61b-8	-19	62b-8	-4	63b-8	-16	64b-8	11
61b-9	-10	62b-9	21	63b-9	1	64b-9	-9
61b-10	-12	62b-10	13	63b-10	7	64b-10	0
61b-11	21	62b-11	-7	63b-11	-5	64b-11	3
61b-12	-3	62b-12	13	63b-12	10	64b-12	-2

65a-1	10	66a-1	-6	67a-1	-13	68a-1	13
65a-2	-3	66a-2	17	67a-2	-34	68a-2	13
65a-3	-8	66a-3	6	67a-3	28	68a-3	0
65a-4	1	66a-4	-1	67a-4	-19	68a-4	-18
65a-5	-6	66a-5	9	67a-5	-11	68a-5	-26
65a-6	11	66a-6	-1	67a-6	-4	68a-6	-24
65a-7	7	66a-7	-12	67a-7	-29	68a-7	-29
65a-8	-11	66a-8	5	67a-8	23	68a-8	-19
65a-9	1	66a-9	7	67a-9	-33	68a-9	-5
65a-10	-14	66a-10	-4	67a-10	20	68a-10	-4
65a-11	-1	66a-11	-5	67a-11	16	68a-11	-15
65a-12	-11	66a-12	-2	67a-12	24	68a-12	-5
65b-1	-8	66b-1	-10	67b-1	-3	68b-1	1
65b-2	12	66b-2	-6	67b-2	0	68b-2	-7
65b-3	2	66b-3	-3	67b-3	-20	68b-3	-6
65b-4	-5	66b-4	12	67b-4	-28	68b-4	15
65b-5	6	66b-5	-4	67b-5	1	68b-5	-26
65b-6	-17	66b-6	3	67b-6	-34	68b-6	22
65b-7	1	66b-7	-23	67b-7	-20	68b-7	-24
65b-8	9	66b-8	-6	67b-8	14	68b-8	-32
65b-9	-12	66b-9	12	67b-9	-9	68b-9	-31
65b-10	-2	66b-10	6	67b-10	-14	68b-10	10
65b-11	-6	66b-11	-5	67b-11	25	68b-11	-15
65b-12	4	66b-12	15	67b-12	-12	68b-12	2

69a-1	19	70a-1	5	71a-1	26	72a-1	-1
69a-2	31	70a-2	3	71a-2	21	72a-2	-9
69a-3	-8	70a-3	18	71a-3	-14	72a-3	-17
69a-4	2	70a-4	-15	71a-4	10	72a-4	-16
69a-5	13	70a-5	-4	71a-5	14	72a-5	-6
69a-6	30	70a-6	12	71a-6	5	72a-6	-14
69a-7	6	70a-7	-14	71a-7	-11	72a-7	19
69a-8	-21	70a-8	21	71a-8	-12	72a-8	-11
69a-9	10	70a-9	14	71a-9	5	72a-9	16
69a-10	-1	70a-10	19	71a-10	-8	72a-10	18
69a-11	-18	70a-11	5	71a-11	-10	72a-11	6
69a-12	27	70a-12	10	71a-12	28	72a-12	-5
69b-1	23	70b-1	-12	71b-1	-22	72b-1	2
69b-2	-16	70b-2	-9	71b-2	-3	72b-2	-17
69b-3	28	70b-3	21	71b-3	-9	72b-3	-19
69b-4	22	70b-4	7	71b-4	-16	72b-4	-12
69b-5	23	70b-5	-7	71b-5	3	72b-5	-5
69b-6	10	70b-6	-11	71b-6	14	72b-6	-3
69b-7	17	70b-7	4	71b-7	11	72b-7	-14
69b-8	18	70b-8	17	71b-8	7	72b-8	1
69b-9	-23	70b-9	-7	71b-9	7	72b-9	-7
69b-10	-5	70b-10	-1	71b-10	-3	72b-10	4
69b-11	5	70b-11	16	71b-11	8	72b-11	17
69b-12	11	70b-12	-22	71b-12	0	72b-12	-24

73a-1	16	74a-1	-7	75a-1	11	76a-1	-15
73a-2	-45	74a-2	36	75a-2	-26	76a-2	-28
73a-3	-46	74a-3	-2	75a-3	39	76a-3	19
73a-4	15	74a-4	-5	75a-4	-52	76a-4	17
73a-5	-27	74a-5	-14	75a-5	1	76a-5	-22
73a-6	16	74a-6	-5	75a-6	-9	76a-6	22
73a-7	1	74a-7	14	75a-7	10	76a-7	-27
73a-8	8	74a-8	33	75a-8	26	76a-8	9
73a-9	25	74a-9	-41	75a-9	-47	76a-9	-8
73a-10	34	74a-10	-48	75a-10	-5	76a-10	7
73a-11	-7	74a-11	11	75a-11	31	76a-11	-19
73a-12	7	74a-12	-2	75a-12	-16	76a-12	30
73b-1	18	74b-1	5	75b-1	-38	76b-1	-22
73b-2	44	74b-2	1	75b-2	-33	76b-2	-9
73b-3	1	74b-3	54	75b-3	-47	76b-3	21
73b-4	-5	74b-4	33	75b-4	21	76b-4	-18
73b-5	-13	74b-5	1	75b-5	-15	76b-5	23
73b-6	33	74b-6	16	75b-6	8	76b-6	18
73b-7	-31	74b-7	36	75b-7	-15	76b-7	-32
73b-8	1	74b-8	9	75b-8	-5	76b-8	6
73b-9	-2	74b-9	40	75b-9	14	76b-9	-33
73b-10	-37	74b-10	31	75b-10	-6	76b-10	7
73b-11	9	74b-11	-21	75b-11	-1	76b-11	-39
73b-12	14	74b-12	26	75b-12	-5	76b-12	-12

77a-1	33	78a-1	-5	79a-1	-40	80a-1	66
77a-2	-30	78a-2	-22	79a-2	-10	80a-2	27
77a-3	21	78a-3	26	79a-3	1	80a-3	-7
77a-4	-5	78a-4	7	79a-4	-60	80a-4	17
77a-5	11	78a-5	-25	79a-5	44	80a-5	-14
77a-6	39	78a-6	-19	79a-6	-21	80a-6	-30
77a-7	21	78a-7	-17	79a-7	-17	80a-7	-38
77a-8	-46	78a-8	3	79a-8	36	80a-8	-36
77a-9	13	78a-9	-33	79a-9	3	80a-9	15
77a-10	24	78a-10	28	79a-10	42	80a-10	-18
77a-11	-18	78a-11	29	79a-11	-8	80a-11	2
77a-12	-22	78a-12	-25	79a-12	17	80a-12	-6
77b-1	-39	78b-1	19	79b-1	-42	80b-1	17
77b-2	-5	78b-2	-34	79b-2	-22	80b-2	41
77b-3	21	78b-3	1	79b-3	1	80b-3	-52
77b-4	1	78b-4	-32	79b-4	18	80b-4	44
77b-5	24	78b-5	49	79b-5	57	80b-5	-34
77b-6	17	78b-6	-20	79b-6	-30	80b-6	-29
77b-7	39	78b-7	-10	79b-7	-39	80b-7	-13
77b-8	29	78b-8	50	79b-8	3	80b-8	-25
77b-9	-1	78b-9	21	79b-9	-72	80b-9	0
77b-10	-14	78b-10	19	79b-10	-78	80b-10	10
77b-11	-14	78b-11	13	79b-11	0	80b-11	9
77b-12	-2	78b-12	29	79b-12	10	80b-12	-4

81a-1	18	82a-1	-4	83a-1	-27	84a-1	31
81a-2	2	82a-2	4	83a-2	-8	84a-2	14
81a-3	-26	82a-3	7	83a-3	24	84a-3	21
81a-4	-50	82a-4	16	83a-4	-24	84a-4	15
81a-5	-32	82a-5	5	83a-5	73	84a-5	-18
81a-6	-36	82a-6	14	83a-6	75	84a-6	-24
81a-7	12	82a-7	-6	83a-7	18	84a-7	69
81a-8	19	82a-8	-40	83a-8	34	84a-8	23
81a-9	-6	82a-9	-11	83a-9	-41	84a-9	-77
81a-10	3	82a-10	22	83a-10	37	84a-10	-36
81a-11	19	82a-11	26	83a-11	10	84a-11	-9
81a-12	-68	82a-12	52	83a-12	50	84a-12	4
81b-1	32	82b-1	-12	83b-1	-37	84b-1	0
81b-2	33	82b-2	-13	83b-2	19	84b-2	-5
81b-3	7	82b-3	-14	83b-3	1	84b-3	-15
81b-4	45	82b-4	15	83b-4	47	84b-4	8
81b-5	-8	82b-5	-8	83b-5	75	84b-5	30
81b-6	5	82b-6	-27	83b-6	-2	84b-6	-21
81b-7	-36	82b-7	-55	83b-7	16	84b-7	59
81b-8	23	82b-8	-9	83b-8	-16	84b-8	28
81b-9	25	82b-9	3	83b-9	3	84b-9	22
81b-10	39	82b-10	-5	83b-10	-20	84b-10	-12
81b-11	-21	82b-11	3	83b-11	44	84b-11	27
81b-12	-8	82b-12	50	83b-12	4	84b-12	-36

85a-1	-26	86a-1	-10	87a-1	1	88a-1	7
85a-2	22	86a-2	43	87a-2	-24	88a-2	49
85a-3	-2	86a-3	37	87a-3	-14	88a-3	-42
85a-4	-22	86a-4	40	87a-4	-38	88a-4	2
85a-5	8	86a-5	1	87a-5	-36	88a-5	68
85a-6	7	86a-6	-54	87a-6	-10	88a-6	12
85a-7	-39	86a-7	8	87a-7	76	88a-7	-22
85a-8	6	86a-8	24	87a-8	44	88a-8	-39
85a-9	-50	86a-9	23	87a-9	-1	88a-9	-63
85a-10	6	86a-10	-59	87a-10	-66	88a-10	-18
85a-11	16	86a-11	-32	87a-11	-18	88a-11	62
85a-12	16	86a-12	37	87a-12	46	88a-12	7
85b-1	-27	86b-1	21	87b-1	25	88b-1	21
85b-2	51	86b-2	23	87b-2	-1	88b-2	16
85b-3	63	86b-3	-68	87b-3	14	88b-3	-15
85b-4	-6	86b-4	-51	87b-4	1	88b-4	74
85b-5	63	86b-5	14	87b-5	0	88b-5	-25
85b-6	29	86b-6	-55	87b-6	48	88b-6	-7
85b-7	-14	86b-7	-13	87b-7	-28	88b-7	-28
85b-8	29	86b-8	23	87b-8	-65	88b-8	-9
85b-9	-17	86b-9	-18	87b-9	-4	88b-9	27
85b-10	32	86b-10	-3	87b-10	12	88b-10	-54
85b-11	-6	86b-11	46	87b-11	29	88b-11	-12
85b-12	-22	86b-12	-54	87b-12	-44	88b-12	23

89a-1	-21	90a-1	16
89a-2	44	90a-2	-18
89a-3	-72	90a-3	7
89a-4	24	90a-4	5
89a-5	-16	90a-5	6
89a-6	6	90a-6	13
89a-7	33	90a-7	20
89a-8	45	90a-8	20
89a-9	33	90a-9	-60
89a-10	45	90a-10	55
89a-11	-33	90a-11	-26
89a-12	55	90a-12	-48
89b-1	-53	90b-1	-9
89b-2	26	90b-2	8
89b-3	20	90b-3	-24
89b-4	1	90b-4	-13
89b-5	-47	90b-5	33
89b-6	1	90b-6	44
89b-7	58	90b-7	36
89b-8	-12	90b-8	-17
89b-9	15	90b-9	71
89b-10	-1	90b-10	25
89b-11	13	90b-11	-11
89b-12	-41	90b-12	-23